

REVIEW

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Asking the classroom with technology: a systematic literature review

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Abstract

Response technologies (RTs), also termed clickers or student response systems, have gained traction among researchers in classrooms in recent years. RTs encompass various interactive tools and technologies that are pivotal in modern educational settings. Numerous articles emphasize the effectiveness of RTs across multiple grades and courses. Existing literature reviews on the subject need a systematic review from a school classroom perspective. This article aims to present a systematic review of the literature on the effectiveness of RTs in school classrooms and highlight future guidelines for implementing RTs. We conducted a systematic literature review, collecting 105 peer-reviewed studies covering different school grades and courses. Our work presented the mapping of RTs, a graphical summary of our work, the applications of RTs in major courses at primary, middle, and high school levels, and future guidelines for implementing RTs in school classrooms under one umbrella. A significant number of studies reported positive learning outcomes. Still, few studies reported insignificant learning outcomes due to the digital divide, deficiency of proper infrastructure, limited support for innovative practices, lack of compatibility with other existing platforms, and inadequate digital literacy. This study provides recommendations for technologists (human-centered design, dynamic customization, advanced data protection protocols, cross-platform adaptability), teachers (digital competence, lecture planning, peer collaboration, adaptive teaching), institutions (resources allocation, support of interactive practices, impact monitoring, incentives for teachers) policymakers (training programs, equitable access, funding, and resources, clear guidelines for implementation) for successful implementation.

Keywords: Classrooms, Educational technology, Interactive lectures, Response technology, Student response systems

Introduction

RTs imply a range of digital tools or artifacts that aid interactions and feedback among students and teachers in the classroom through the Internet (Crouch & Mazur, 2001). Earlier, RTs were perceived solely as hardware devices because they originated with separate units like clickers (Judson & Sawada, 2002). Modern RTs have evolved to incorporate various tools, frequently through Bring Your Own Device, which relies on software and internet connectivity (Chou, 2017). RT then covers a plethora of systems under various names, including classroom response systems, student response systems, audience

response systems, electronic response systems, or clickers (Judson & Sawada, 2002; Menon et al., 2004; Stowell et al., 2010). Since 1966, RTs have advanced with hand-held devices, although they have become independent of explicit hardware with technological innovation (Binek et al., 2016; Watson & Berry, 2022). High-tech developments in the past two decades have directed the growth of these technologies, encouraging interaction in synchronous modes (Tarazi & Ortega-Martín, 2023). It boosts social skills, motivation (Purba et al., 2019; Stoyanova et al., 2017), engagement, and reduced anxiety (Turan & Meral, 2018), which leads to a better learning environment for students (Chien et al., 2015). Various articles have revealed that the traditional approach of lectures in the classroom decreased learning motivation (Compton & Allen, 2018), where students passively absorb the lecture from the teacher (Sun & Hsieh, 2018). Several instructional approaches (gamified, flipped learning, etc.) have recently focused on these challenges, highlighting student-centered learning (Stoyanova et al., 2017; Sun & Lin, 2022). These approaches aim to advance interactive and collaborative learning environments (Yanuarito et al., 2023). The application of RTs in school classrooms is one of the fascinating developments over the past few decades. Schools increasingly integrate RTs to enhance the learning experience and prepare students for future endeavors (Agarwal et al., 2014). However, many schools still hesitate to adopt these technologies as interventional classroom tools (Barrio et al., 2015). These technologies assist teachers in posing questions for better interaction and provide immediate feedback during lectures (Harahap & Kembaren, 2023). These are acquired in numerous school grades and are adaptable from small groups to larger classes (Barth-Cohen et al., 2016). The articles in the past decade indicated a significant rise in RTs in school classrooms. Hand-held clickers evolve into web-based tools (Malayas et al., 2022), non-gamified RTs streamline participation (Compton & Allen, 2018), and gamified RTs make an enjoyable learning (Zainuddin et al., 2020). As traditional clickers decline, gamified RTs are expected to dominate, improving real-time interaction with fun and enjoyment (Tsihouridis et al., 2018). RTs are projected to strengthen dynamic, tech-driven pedagogical approaches for modern learners (Ghanaat Pisheh et al., 2019). Using RTs in large classrooms is beneficial in addressing barriers to class participation (Malandrino et al., 2014), personalized feedback (Pérez-Segura et al., 2022), and enhancing the learning experience (Sun & Hsieh, 2018). It allows for anonymous interaction, overcoming communication detention in engaging students (Plump & LaRosa, 2017), break up lectures, refocus students (Sun & Hsieh, 2018), promote active learning (Malandrino et al., 2014), and facilitate peer discussions (Hu & Huang, 2015). RTs facilities better formative and summative assessment that may lead to higher grades (Catindig & Prudente, 2019; Sheard & Chambers, 2014). Traditional methods use paper for student assessment, which is also time-consuming. The use of RTs for student assessment can reduce paper usage to conserve natural resources (Al Munawaroh, 2021; dos Santos Teotônio et al., 2020).

Several researchers have reported that teachers require a significant effort to design interactive lectures (Agbatogun, 2014b; Sinnivasagam & Hua, 2023). Additionally, their use is not guaranteed to enhance overall student performance consistently (Janković & Lambić, 2022). Several studies reported that the use of RTs has no significant impact on students' engagement (Tarazi & Ortega-Martín, 2023), anxiety (Can et al., 2021; España-Delgado, 2023; ÖDEN et al., 2021), motivation (Sun & Hsieh, 2018), knowledge

construction (Sun & Lin, 2022), learning achievement (Maming et al., 2023; Rahmehani, 2020), collaborative learning (Barth-Cohen et al., 2016), summative assessment (Chou et al., 2017), attention (Cahyani et al., 2022), active participation (Cakiroglu et al., 2018), conceptual understanding (Wulandari et al., 2023), and language learning (ÖDEN et al., 2021). These insignificant outcomes may be due to various challenges such as cost issues (Binek et al., 2016; Chien et al., 2015), lack of proper infrastructure (Kartini & Sriyanto, 2023; Shyr et al., 2021), limited support for innovative practices (Ningsih & Mulyono, 2019), the limited functionalities in RTs (Da-Oh et al., 2023; España-Delgado, 2023; Stanojević & Randelović, 2018; Zainuddin et al., 2020), lack of compatibility with other existing platforms (Sun & Hsieh, 2018; Zainuddin et al., 2020), limited digital literacy (Měkota & Marada, 2020; Md Yunus & Azmanuddin bin Azman, 2019), ineffective instructional practices (Agbatogun, 2014b; Cakiroglu et al., 2018), and lack of clear guidelines (Katemba et al., 2022) to implement RTs in classrooms. Teachers used only a few functionalities of RTs due to a lack of digital literacy, leading to only surface-level student engagement (Stanojević & Randelović, 2018; Md Yunus & Azmanuddin bin Azman, 2019). A lack of digital competency may result in technical issues affecting students and teachers during lectures (Měkota & Marada, 2020). Furthermore, teachers sometimes find it challenging to adapt their lesson plans in real time in response to student input via RTs, particularly during lectures (Agbatogun, 2014b). School leaders or administrators must balance RT expenses, technical assistance, and teacher training to ensure successful implementation in learning environments (Binek et al., 2016; Cakiroglu et al., 2018; Shyr et al., 2021). Furthermore, it is challenging for teachers and students to access appropriate personal devices, especially in developing countries where the digital divide has been prevalent (Chien et al., 2015). The technical and lack of infrastructure-related challenges are reported mainly by developing countries such as Malaysia and Indonesia (Kartini & Sriyanto, 2023; Yanuarto et al., 2023) as compared to developed countries due to geographic and economic differences or due to norms in society and cultural patterns of schooling (Gustafsson & Ryve, 2022). This highlights the positive and negative aspects of RTs in classrooms that need further exploration, which leads to the following research questions:

- How are RTs effective in different levels of school classrooms as interventional tools?
- What are the challenges and future guidelines associated with integrating RT in school classrooms?

This systematic literature review (SLR) enables a deeper understanding of the various dimensions and contributions of RT research that span the context of school classrooms.

Evolution of response technologies

Traditional methods of student participation, such as raising hands or verbal responses, are commonly used in classrooms (Compton & Allen, 2018). Electric response systems emerged in the 1970s, originating from the military use of filmed instruction in the 1960s. These systems were electronically advanced for their time, offering anonymous options (Alberts & Papp, 1997). Tagged systems recorded responses from each seat, while anonymous systems provided response counts.

During this period, they were focused on giving students and teachers quick feedback (Judson & Sawada, 2002). During this period, electric response systems were used to allow students to provide feedback on the pace of instruction. Still, different articles concluded that these electric response systems are open to the classroom environment (Littauer, 1972). Communally, research from the 1970s linked to electronic response systems does not establish the claim that student achievement is enhanced with the aid of such systems (Crouch & Mazur, 2001). During the 1980s, RTs advanced in the classroom, from early remote-control systems to more refined wireless technologies. The induction of clickers and adding RTs with computers are concrete steps toward more collaborative and data-driven educational capabilities (Menon et al., 2004). During the 1990s, clickers convulsed to increase popularity and were progressively used in classrooms. The technology behind clickers also advanced during this time. Wireless and radio frequency-based clickers became more frequent, switching earlier infrared-based systems. These developments permitted more straightforward functioning and seamless addition to classroom environments (Campbell & Piccinin, 1999). Professor Eric Mazur from Harvard University has promoted clickers in peer instruction. This fostered students to engage in valuable discussions with their peers during class, aiming at conceptual questions related to the lecture material (Crouch & Mazur, 2001).

In 2002, Stanford University School of Medicine led a personal digital assistant (PDA)-based response system in a histology class, acknowledged students and teachers to present interactive feedback and immediate group responses to questions. In these systems, students utilized PDAs to answer questions and quizzes posed by their teachers. PDAs were small hand-held devices with response software, allowing students to submit answers and engage in classroom activities (Menon et al., 2004). Similarly, in 2005, Ohio State University utilized a wireless keypad voting machine system for in-class polling, enabling students to answer multiple-choice questions during lectures (Reay et al., 2005). The utilization of voting machines enhanced classroom dynamics, particularly fostering student–teacher interactivity. In the late 2000s, the widespread use of mobile phones among students opened new opportunities for classroom interaction systems (Scornavacca et al., 2009). They have highlighted the benefits of short message service (SMS) messages as an additional communication channel in the classroom for better feedback and engagement. However, the price of SMS subscriptions and the high cost of cell phones were primary barriers. Hand-held or traditional clickers implemented more adaptable and technologically advanced options during the 2000s to overcome these challenges. Many new RTs and related tools entered the market in the early 2010s when smartphones and laptops were widely available, facilitating online access and wireless network connectivity (Stowell et al., 2010). In the early 2010s, due to high availability of smartphones and laptops, Web and mobile-based RTs revolutionized education by providing interactive tools for teachers and students (Binek et al., 2016). Furthermore, these tools also provided data analytics to monitor student’s progress to evaluate the effectiveness of pedagogical practices (Chou et al., 2017). The evolution of RTs is presented in Fig. 1.

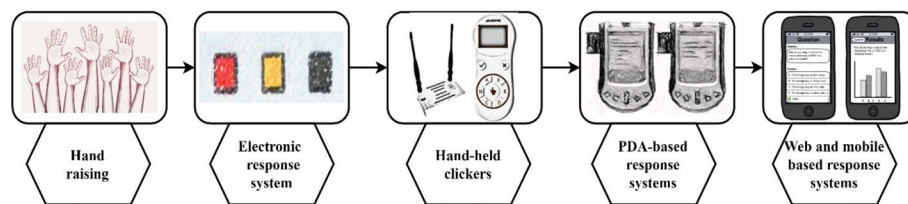


Fig. 1 The evolution of RTs

Contributions of the article

The primary objective is to extensively assess the RTs that find application in school classrooms, shedding light on the prevailing trends in their adoption. To the best of our knowledge, a conspicuous gap exists in the current academic literature about comprehensive surveys that systematically explain the complex utilization of RTs in school classrooms. Our work also presented the mapping of RTs, a graphical summary of our work, the applications of RTs at primary, middle, and high school levels in major courses (science, mathematics, and language courses), trends and divergences, challenges and future guidelines for implementation of RTs in school classrooms under one umbrella.

We have focused on border prescriptive and explored the effectiveness of RTs in school classrooms at different grade levels. Compared to the existing survey articles in the area, our survey is inclusive and unique in providing the following aspects: providing the evolution of RTs, giving descriptions and comparison of various categories of RTs concerning their type, summarizing the analysis of RTs in school classrooms based on study type, sample size, section's count, evaluation method, evaluation metrics, and pointing out emerging challenges. The prevalent themes across many articles were student engagement, learning motivation, learning performance, and challenges. These aspects were consistently addressed, indicating their significance in the articles and suggesting their importance in shaping educational practices and outcomes. A graphical summary of our work's significant components is also provided in Fig. 2.

Organization of the article

Figure 3 illustrates the comprehensive structural layout of the article, which provides a holistic understanding of RTs in school classrooms.

Methodology

This study aims to review the utilization of RTs in school classrooms systematically. A SLR is deemed appropriate for collecting and summarizing articles on the subject, offering a comprehensive perspective (Kitchenham et al., 2009). The methodology includes study selection (databases used, search string construction, inclusion and exclusion criteria), the systematic review process, quality assessment, data extraction, and synthesis.

Study selection

Databases used

A comprehensive search query was established to identify relevant articles within the domain of RTs to find articles from academic databases, namely Web of Science,

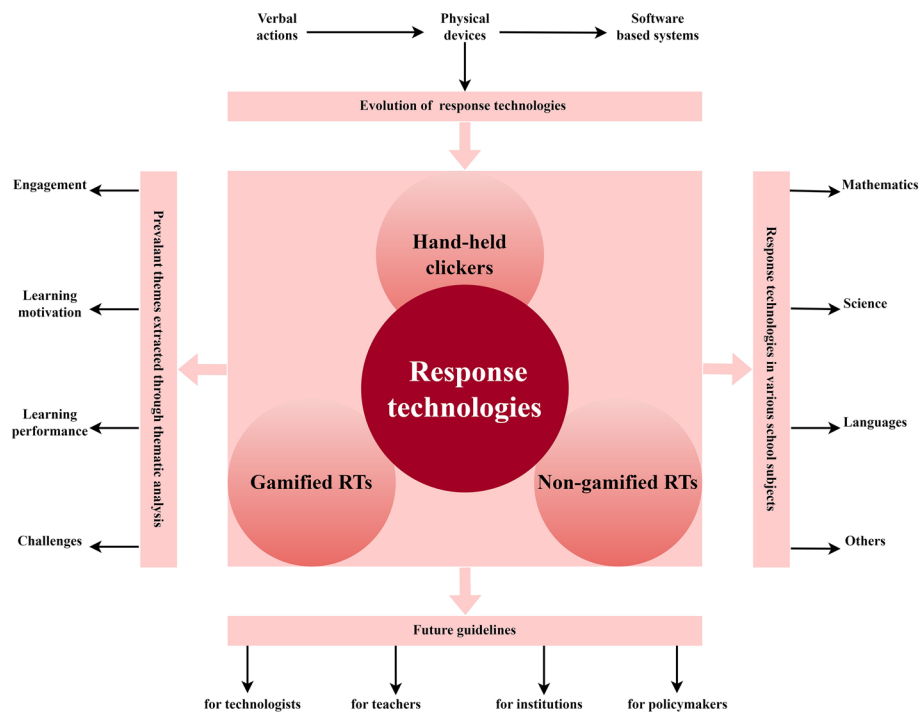


Fig. 2 The visual summary of our work

Scopus, Google Scholar (multidisciplinary databases), ProQuest and ERIC (educational databases), and ACM library (computing and technology database). The selection of these online databases was deliberate, driven by their capacity to provide access to a substantial repository of various journals, conference proceedings, and book chapters that span the expansive domain of RTs and their interplay with school education. Furthermore, the snowball method was also applied to reduce bias by identifying additional relevant studies through references (backward snowballing) or citations (forward snowballing) of initially selected papers. This method broadens the search scope, ensuring comprehensive topic coverage while minimizing the risk of missing key literature. However, some low-quality articles were also selected during the snowballing method. An extensive quality assessment method for full-text articles was applied to resolve this bias after choosing the primary studies to remove the low-quality articles.

Search string construction

The search query was constructed using specific terms and Boolean operators to capture relevant articles. Considering the multidisciplinary domain in the review, keywords were established to represent each study area. A search query was formulated using the two main terms: “response technology” AND “education”. Potential synonyms were included after reviewing the related research from the initial search. After refining several variations of these key terms, the resulting query was used to conduct our search. The search string was (“student* response system*” OR “learner* response system*” OR “audience response system*” OR “personal response system*” OR “classroom* response system*” OR “class* response system*” OR “interactive

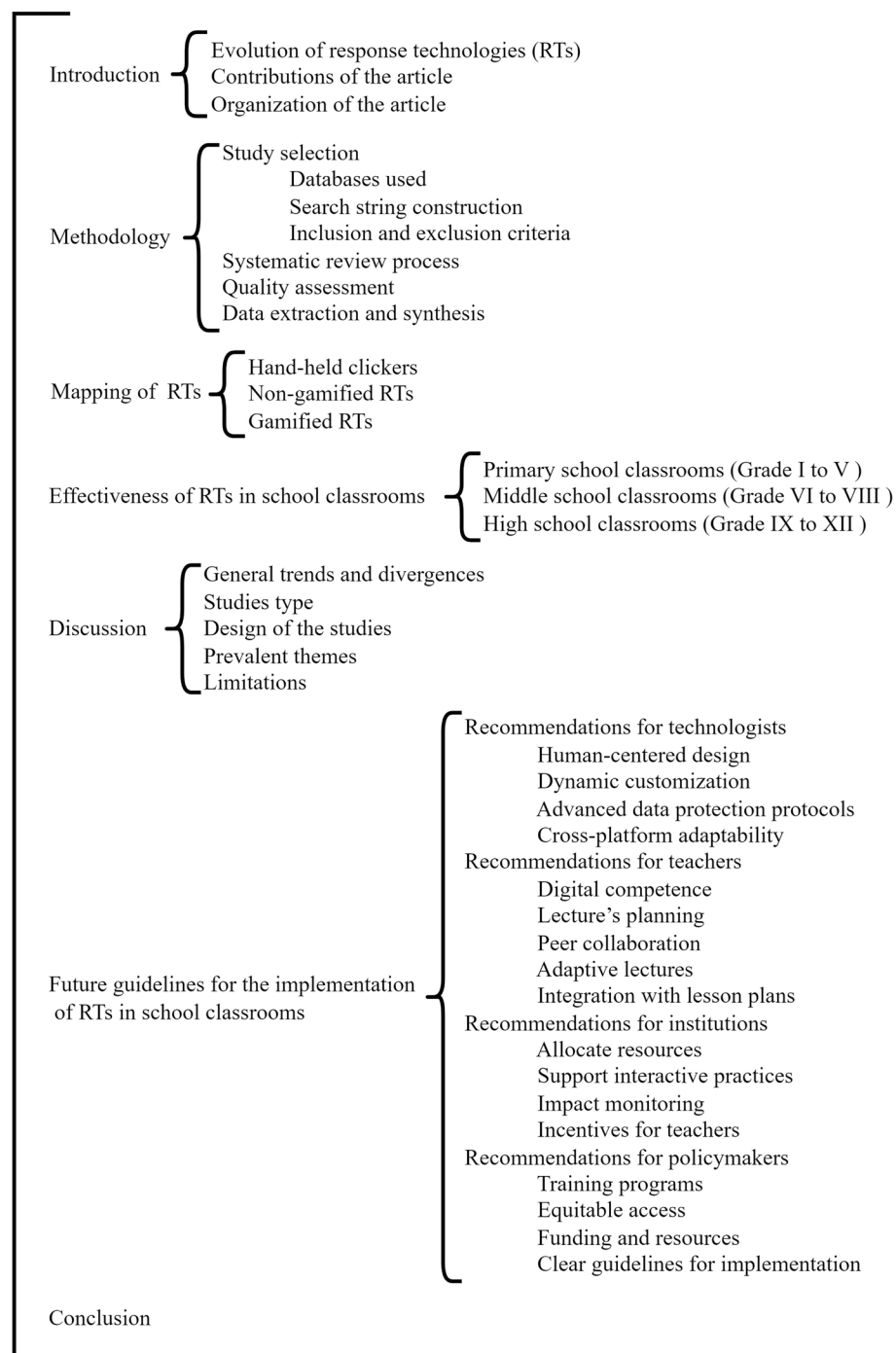


Fig. 3 Structural layout of our article

response system*" OR "instant response system*" OR "online response system*" OR "web-based response system*" OR "game*-based response system*" OR "electronic* response system*" OR "clicker*" OR "physical* AND clickers*" OR "electronic voting system*" OR "interactive voting system*" OR "classroom communication system*" OR "classroom performance system*" OR "electronic feedback system*" OR "polling

system*" OR "voting system*" OR "response technolog*") AND (classroom OR education OR educational OR instruction OR instructional OR learning OR peer OR promotive* OR pupil OR school OR small-group OR student OR teacher OR teaching OR whole-class), in the title, abstract, and keywords. The search query focused on the retrieved articles' titles, abstracts, and keywords. Furthermore, to ensure persistence and penetration of the search, Google Scholar was employed to cross-reference the results, as well as forward and backward searching to retrieve less common libraries.

Inclusion and exclusion criteria

Our SLR ranged from 2014–2023 based on several considerations, such as the widespread use of smartphones and tablets, technological advancements in RTs, and increasing research focusing on these RT's effectiveness, implementation, and pedagogical implications. The articles published before 2014 may offer valuable insights into the broader context of educational technology. Additionally, papers focusing on RT using empirical data and RTs in schools were considered. Non-RT studies, theoretical studies that are not peer-reviewed or gray literature, and unavailable full-text papers are among the exclusion criteria. Table 1 displays the inclusion and exclusion criteria.

Systematic review process

A thorough set of inclusion and exclusion criteria was applied to find articles that align with the research objectives. The 4817 articles were imported into EndNote, a reference management software program. While checking for duplicates, 2249 duplicate records were removed. The first two authors meticulously assessed the relevance of all studies by scrutinizing their titles and abstracts, eliminating irrelevant papers unrelated to research questions. The article inclusion process was iterative, marked by regular consensus meetings between the first two authors. These meetings centered on crucial variables guiding the unbiased and ethical analysis of selected papers to address the research questions. The titles and abstracts of 2568 articles were screened aimed at discerning potential articles that aligned with the objectives of our study. Of these, 608 underwent a thorough full-text evaluation, adhering strictly to the predefined inclusion criteria. From this pool, 512 articles were excluded, guided by specific exclusion criteria. Additionally, articles that failed to utilize RTs as a critical part of their investigation, articles wherein RTs did not serve as the primary focus. Furthermore, articles categorized as literature reviews were expressly omitted. Most excluded studies were either from higher education, outside of the classroom, or not empirical studies. Through forward and backward

Table 1 Inclusion and exclusion criteria

Inclusion criteria	Exclusion criteria
1. Utilized RTs in school classrooms	1. Articles that do not employ RTs and lack clarity regarding using RTs in school classrooms
2. Published in the English language	2. Materials that are not available as full-text and not peer-reviewed
3. Published within the timeframe from 2014 to December 23, 2023	3. Purely theoretical articles
4. RTs must serve as the central and predominant part of the study	
5. Articles that analyzed the empirical data regarding RTs	

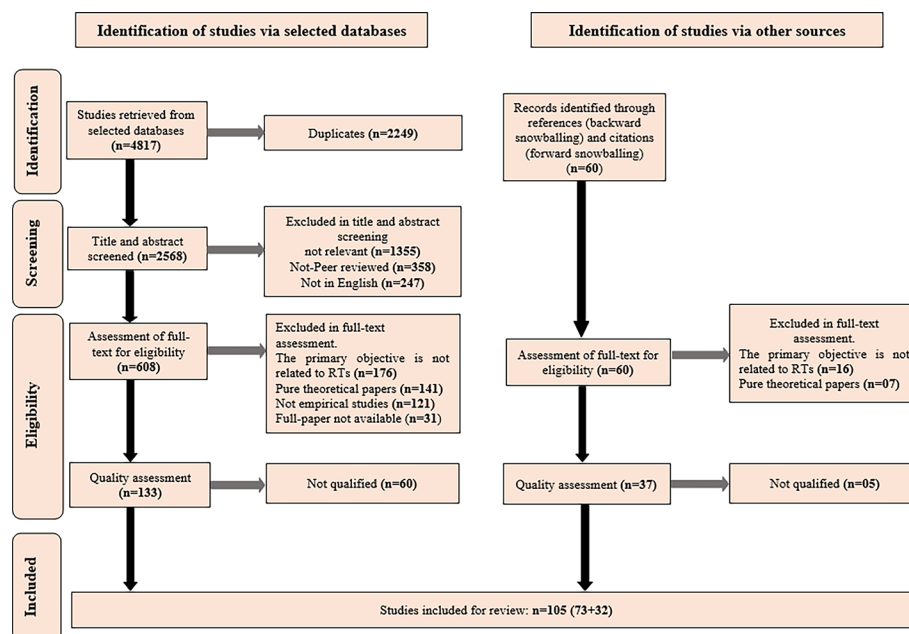


Fig. 4 A visual representation of the article selection process

searches, 37 out of 60 articles were added to a refined selection of 133. An illustrative depiction of the process for selecting articles is presented in Fig. 4.

Quality assessment

To maintain the rigor of our study, we conducted a quality assessment of the included studies. This process aimed to select only high-quality research that directly addressed our research questions and minimize bias in the study selection process. The first two authors assessed each of the 133 studies retained after evaluating through nine quality assessment criteria. These criteria were derived from the Critical Appraisal Skills Programme (CASP) (Dybå & Dingsøyr, 2008), following a methodology employed in previous SLRs on multidisciplinary subjects, such as educational technology, management, and software development (Kitchenham et al., 2009). The following are the quality assessment criteria:

QA1: Is the paper based on empirical research?

QA2: Are the clear statement of the research aims explained?

QA3: Is the description of the research context provided?

QA4: Was the research design appropriate to address the research aims?

QA5: Has the author explained the selected participants and elaborated on the number of participants who used RT?

QA6: Is it clear which data collection method/methods were used?

QA7: Was the data analysis appropriately rigorous, with a description of the analysis explained to support the findings?

QA8: Are the findings explicit enough to support the discussion and possible limitations?

QA9: Do the researchers discuss the contributions and mention future work?

Table 2 Quality assessment of the selected articles

Reference	QA1	QA2	QA3	QA4	QA5	QA6	QA7	QA8	QA9	Score
Sinnivasagam & Hua, (2023)	✓	✓	✓	✓		✓	✓	✓	✓	8
Harahap & Kembaren, (2023)	✓	✓		✓			✓	✓	✓	6
Janković et al., (2023)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Yanuarto et al., (2023)	✓	✓	✓			✓	✓	✓	✓	7
Cahyaningtyas & Chakim, (2023)	✓	✓			✓	✓		✓	✓	6
Watson & Berry, (2022)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Abdullah et al., (2022)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Janković & Lambić, (2022)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Mahmud & Law, (2022)	✓	✓	✓	✓		✓	✓			6
Sartono et al., (2021)	✓	✓	✓	✓	✓	✓	✓			7
Dociu & Mihut-Kert, (2021)	✓	✓	✓	✓	✓	✓	✓		✓	8
Fiani et al., (2021)	✓	✓	✓	✓	✓	✓	✓			7
Pratama, (2021)	✓	✓	✓	✓	✓	✓	✓			7
Siwalette, (2021)	✓	✓	✓		✓		✓			5
Wulandari & Hartono, (2021)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Anak Yunus & Hua, (2021)	✓	✓	✓	✓	✓	✓	✓			7
Djannah, (2021)	✓	✓	✓	✓	✓	✓				6
Umboh et al., (2021)	✓	✓	✓	✓		✓				5
Halim et al., (2020)	✓	✓	✓	✓		✓	✓	✓		7
Idris et al., (2020)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Setiawan & Soeharto, (2020)	✓	✓	✓	✓	✓	✓	✓		✓	8
Mékota & Marada, (2020)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Sukowati & Sartono, (2020)	✓	✓	✓	✓	✓	✓	✓	✓		8
Wong & Yunus, (2020)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Lu et al., (2019)	✓	✓	✓	✓	✓	✓				6
Ningsih & Mulyono, (2019)	✓	✓	✓			✓	✓	✓		6
Hakim et al., (2019)	✓	✓	✓	✓	✓		✓			6
Md Yunus & Azmanuddin bin Azman (2019)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Stanojević & Randelović (2018)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Cakiroglu et al. (2018)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Hou (2018)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Chou (2017)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Agbatogun (2014a)	✓	✓	✓	✓		✓	✓	✓	✓	8
Agbatogun (2014b)	✓	✓	✓	✓		✓	✓	✓	✓	8
Sheard & Chambers (2014)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Balaskas et al. (2023)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
España-Delgado (2023)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Kartini& Sriyanto (2023)	✓	✓	✓	✓	✓	✓	✓	✓		8
Nordin (2023)	✓	✓	✓	✓		✓	✓	✓	✓	8
Da-Oh et al. (2023)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Wulandari et al. (2023)	✓	✓	✓	✓		✓	✓			6
Koç & Sütçü (2023)	✓	✓	✓	✓	✓	✓	✓		✓	8
Sun & Lin (2022)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Rusmardiana et al. (2022)	✓	✓	✓	✓	✓	✓	✓		✓	8
Gustafsson & Ryve (2022)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Rosyida et al. (2022)	✓	✓	✓	✓	✓	✓				6
Sénécal et al. (2022)	✓	✓	✓	✓		✓				5
Pérez-Segura et al. (2022)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Katamba et al. (2022)	✓	✓	✓	✓		✓	✓	✓	✓	8

Table 2 (continued)

Reference	QA1	QA2	QA3	QA4	QA5	QA6	QA7	QA8	QA9	Score
Malayas et al. (2022)	✓	✓	✓	✓		✓				5
Shyr et al. (2021)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Rochimah & Muslim (2021)	✓	✓	✓	✓		✓	✓			6
Baguio et al. (2021)	✓	✓	✓		✓	✓	✓		✓	7
Obery et al. (2021)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Zainuddin et al. (2020)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Mays et al. (2020)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Curto Prieto et al. (2019)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Ghanaat Pisheh et al. (2019)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Zhu & Urhahne (2018)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Videnovik et al. (2018)	✓	✓				✓	✓	✓	✓	6
Sun & Hsieh (2018)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Turan & Meral (2018)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Tsihouridis et al. (2018)	✓	✓	✓	✓		✓	✓	✓	✓	8
Chou et al. (2017)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Cohn & Fraser (2016)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Barth-Cohen et al. (2016)	✓	✓	✓	✓		✓	✓	✓	✓	8
Siswati et al. (2023)	✓	✓	✓	✓		✓	✓	✓		7
Xin & Johnson (2015)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Lee et al. (2019)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Harefa et al. (2020)	✓	✓	✓	✓		✓	✓		✓	7
Yana et al. (2022)	✓	✓	✓	✓	✓	✓	✓			7
Hu & Huang (2015)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Maming et al. (2023)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Tarazi & Ortega-Martín (2023)	✓	✓	✓	✓			✓	✓	✓	7
Powa & Murniarti (2022)	✓	✓		✓	✓		✓			5
Buele-Guerrero & Torres-Carrión (2022)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Naumoska et al. (2022)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Cahyani et al. (2022)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Suryani & Fauziati (2022)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
ÖDEN et al. (2021)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Can et al. (2021)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Akhsani & Yanuarto (2021)	✓	✓	✓			✓	✓			5
Anh et al. (2021)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Katamba & Sinuhaji (2021)	✓	✓	✓	✓		✓	✓	✓	✓	8
Hikmah et al. (2021)	✓	✓	✓	✓	✓	✓	✓			7
Quiroz et al. (2021)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Al Munawaroh (2021)	✓	✓	✓	✓	✓	✓	✓	✓		8
Malik et al. (2021)	✓	✓	✓	✓		✓	✓		✓	7
Dhamayanti (2021)	✓	✓	✓	✓	✓	✓	✓	✓		8
Rahmahani (2020)	✓	✓	✓	✓		✓	✓	✓		7
Chiang, (2020)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Rahmawati et al. (2020)	✓	✓	✓	✓		✓	✓	✓		7
Setiyani et al. (2020)	✓	✓	✓	✓	✓	✓	✓		✓	8
dos Santos Teotônio et al. (2020)	✓	✓	✓	✓	✓	✓	✓			7
Catindig & Prudente (2019)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Mansur & Fadhilawati (2019)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Purba et al. (2019)	✓	✓	✓	✓	✓	✓		✓		7
Jones et al. (2019)	✓	✓	✓	✓		✓	✓	✓	✓	8

Table 2 (continued)

Reference	QA1	QA2	QA3	QA4	QA5	QA6	QA7	QA8	QA9	Score
Stoyanova et al. (2017)	✓	✓	✓	✓	✓	✓	✓			7
Binek et al. (2016)	✓	✓	✓	✓	✓	✓	✓			7
Barrio et al. (2015)	✓	✓	✓	✓		✓	✓	✓	✓	8
Chien et al. (2015)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Agarwal et al. (2014)	✓	✓	✓	✓		✓	✓	✓	✓	8
Wang et al. (2014)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9
Malandrino et al. (2014)	✓	✓	✓	✓	✓	✓	✓	✓	✓	9

Table 3 Data extraction form

Reference	Author, title, and year.
Study design	Experimental, case study, action, or other methods.
Classroom level	Primary, middle, or high.
Grade	Grade I–V (primary), Grade VI–VIII (middle), Grade IX–XII (high)
Course or Subject	Language, mathematics, science, or other courses.
Type of RTs	Hand-held clickers, non-gamified RTs, or gamified RTs.
Sample size	Number of participants, either students or teachers or both. Only those participants mentioned that they used any RT, i.e., participants from the experimental group, not the control group.
Section's count	It specifies the number of classes or sections in which RTs are utilized.
Evaluation method	The type of evaluation instrument used to measure the effectiveness such as quantitative (questionnaire or content-based assessment), qualitative (interview, observation or field notes) method or both.
Evaluation metrics	Mean, average, percentage, standard deviation etc., (descriptive statistical analysis), t-tests, regression analysis, chi-square tests etc., (inferential statistical analysis), Thematic analysis
Results	What are the findings of the study?

Following our quality assessment, 105 studies were deemed suitable for inclusion in the analysis. We used a binary scoring system to evaluate the quality of each study. Studies with a quality score of 50 percent or higher were considered to meet our standards for inclusion in this SLR, as reported in Table 2.

Data extraction and synthesis

In this phase of the SLR, 105 primary studies were utilized for data extraction form through a structured framework for capturing comprehensive details from each study, as presented in Table 3. The article's basic information, such as author(s), year, and title, was obtained during the analysis process. This form facilitated the recording of crucial details and ensured consistency in the extraction process across all included studies. We focused on gathering relevant information that directly addressed the study's objectives. Following data extraction, a qualitative analysis of the collected studies was conducted using NVivo, a qualitative data analysis computer software package, and tables were created to present the synthesized findings. A method developed by Cruzes and Dyba (Cruzes & Dyba, 2011) was chosen as the methodological approach for synthesizing the results. The types of RTs were coded into handheld clickers, non-gamified, and gamified RTs. The education level was coded into primary

(I–V), middle (VI–VIII), and high (IX–XII) school grades. However, the specific grade concerning education level may differ from country to country. The course or subject at which the study was conducted was also coded. In some cases, the study was conducted in more than one course, and then only those courses were coded in which RT intervened. For example [(Rosyida et al., 2022)], the study was conducted in two courses, but one class was control, and the other was experimental; in these cases, only the experimental course was coded. Furthermore, courses such as English (grammar, comprehension, writing, listening, etc.) and science (general, chemistry, biology, physics) were coded into respective categories. Similarly, the sample size was coded to include participants, either students or teachers, who have used the RT in the classroom. The study design was coded into experimental, quasi-experimental, descriptive, case study, cross-sectional, mixed-methods, action, pilot, comparative, exploratory, and developmental. A few studies have not explicitly mentioned the study design, but it has been coded according to (Oates et al., 2022) design method with the authors' consensus. The section's count or number of classes in which the study was conducted was coded as reported by the article authors, but only those sections were coded in which RT was utilized. In a few articles, the student's grade, course name, or number of sections were missing and stated as Not mentioned (NM). The evaluation method for effectiveness assessment was coded into quantitative (questionnaires or content-based assessment) and qualitative (interviews, observations, or field notes). The evaluation metrics were coded into descriptive, inferential statistical analysis, and thematic analysis. The critical findings examined in the articles were determined to answer the research question.

Mapping of response technologies

In the early years, basic RTs were introduced, typically using electronic and wireless technologies and, later, physical or hand-held clickers. These systems involved hand-held devices, often transmitters, through which students engaged with the teacher's multiple-choice or true/false questions using software like PowerPoint (Crouch & Mazur, 2001). The evolution from hand-held clickers to audience response systems or non-gamified RTs marks a paradigm shift in educational technology (Mays et al., 2020).

The rise of non-gamified RTs transformed this landscape and leveraged ubiquitous digital platforms for real-time student–teacher interaction (Compton & Allen, 2018; Sun & Lin, 2022). Due to technological advancements, software-based RTs emerged, allowing more sophisticated interactions with school students in the classroom (Stowell et al., 2010). Following non-gamified RTs, gamified RTs have developed as a progressive educational tool. The deliberate integration of gamification displays a promise to leverage innovative approaches to continue student interest, highlighting the continuing evolution of RTs to meet the transforming demands of modern educational practices (Rahmehani, 2020; Setiyani et al., 2020). The mapping of RTs is presented in Fig. 5. The following sections will explain the mapping of RTs employed in school classrooms.

Hand-held clickers

Hand-held clickers are one of the RTs that employ infrared signals to transmit data concerning students and a teacher. These technologies have developed from costly,

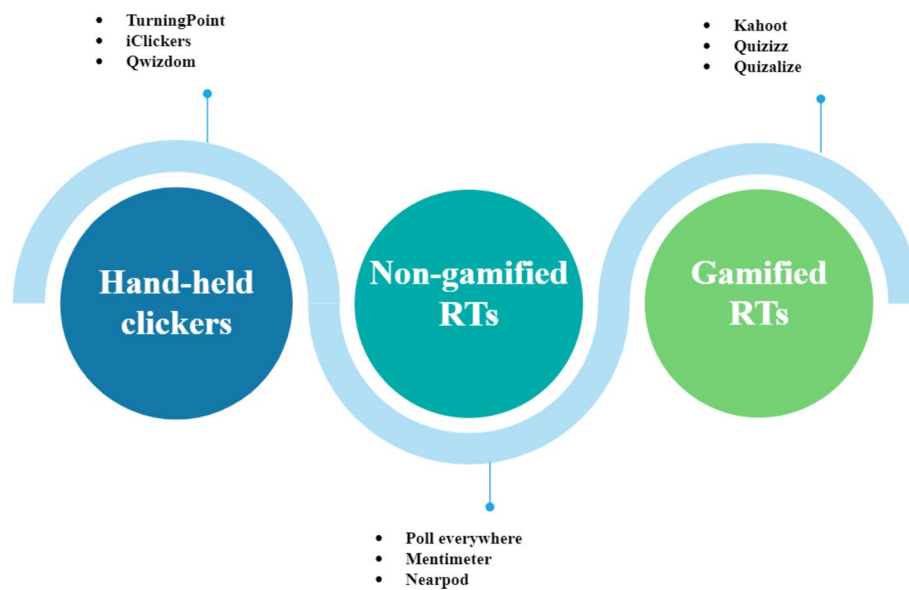


Fig. 5 Mapping of RTs in school classroom settings

location-specific wired setups to cheap, portable systems operating in various settings (Judson & Sawada, 2002). Students use small hand-held transmitters similar to cell phones to transmit their responses to the teacher's receiver via a keypad (Stowell et al., 2010). The teacher's receiver collects these responses and sends them to a computer for real-time processing to view the answers provided by the students. These devices are employed in diverse settings, ranging from large classrooms to smaller discussion-based classes, tutorials, and laboratory sessions (Crouch & Mazur, 2001). In school classrooms, these devices are predominantly acquired from either Turning Technologies, iClickers, or Qwizdom (Echo360, 2023; iClicker, 2023; Qwizdom; Clever, 2023) etc.

Turning Technologies introduced their compact radio frequency keypad, Response-Card, in June 2005. The keypad is seamlessly integrated with PowerPoint through the TurningPoint system, allowing easy question entry and automatic answer choice text boxes. With a small size button, the keypad lacks a decimal point or numeric answer input but features a green light emitting diode light for confirmation (Echo360, 2023). Later, a similar device, iClicker, created by University of Illinois physicists, offered a simple and user-friendly system. The floating menu bar provides easy accessibility across applications. The console-style receiver includes a liquid crystal display for previewing student responses. Teachers may utilize their keypad as a remote control, allowing more freedom. However, this feature may improve the usefulness of displayed student responses for teachers using wireless tablets as remote controls (iClicker, 2023). Concurrently, Qwizdom introduced radio frequency keypads that started with the Q4 model. Qwizdom's proprietary slide design and interactivity allowed the import of pictures. Keypads are registered in a class by entering student and class IDs, with multiple types uses requiring repeated entry of the class ID (Clever, 2023). Hand-held clickers differ in functionality and cost. Some accept only

Table 4 Functionality-based comparison of various hand-held clickers (referred to Clever, 2023; Echo360, 2023; iClicker, 2023))

	Turningpoint	iClicker 2 Remote	Qwizdom QVote
Dimensions (cm; WxHxD)	8.6 × 5.3 × 1.3	5.72 × 12.07 × 1.91	6 × 13.2 × 2.5
Weight (Grams)	25	113	–
Cost (\$)	30–50	40–60	60–80
Batteries	2 Lithium Metal batteries	Two AAA batteries	Two AA batteries
Average battery life	6–12 months	200 h	12 months
Low battery signal	×	✓	×
Technology	RF	RF and Bluetooth Low Energy	RF
Range (feet)	200	200 to 300	50 to 100
Participant registration	×	✓	✓
Number of buttons	12	13	21
Keypad options	0/A – 9/J	A-E	0/A – 9/F
Input options	Alphanumeric	Alphabetic	Alphanumeric
LCD	Small	Small	Large
Image-based questions	×	✓	✓
Automatic Attendance	×	✓	×
Displays right/wrong feedback	×	✓	✓
Options ranking	×	×	✓
Robustness and durability	The design was modified, and fewer problems still exist.	It appears robust but auto shuts off after 90 min.	Appears robust
Student ease of learning	Learning is easy, but students must determine whether their response has registered or not.	Easy to learn, and light indicates that the response has been accepted.	Slight learning curve

single alphanumeric input, while others allow multiple inputs, and compatibility with different systems and proprietary cards varies. Generally, clickers with complex input options tend to be pricier than simpler ones. Updated versions of these devices were also launched over time (Compton & Allen, 2018). We compared the different physical clicker devices as described in (Clever, 2023; Echo360, 2023; iClicker, 2023), and they are presented in Table 4. Due to their design limitations, these are primarily used as in-classroom response devices and are unsuitable for remote or online use. Furthermore, researchers have reported different barriers such as cost issues (Binek et al., 2016; Chien et al., 2015), hesitant to adopt new technologies due to perceived time and effort (Agbatogun, 2014a), and technical issues also leading to frustration (Stowell et al., 2010).

Non-gamified response technologies

The rise of portable devices such as smartphones, tablets, and laptops has led to the development of software-based RTs that allow students to respond using their own connected devices, often named audience response systems, cloud-voting RTs, or non-gamified RTs (Compton & Allen, 2018). In developed countries, where many school students have smartphones, accessing these RTs via their devices is promising (Hou, 2018). This shift has increased the accessibility of RTs and opened opportunities for synchronous



Fig. 6 A generic workflow of RTs in the classroom

interactive lectures beyond traditional lectures (Chou et al., 2017). A pedagogical workflow of RTs during lectures is shown in Fig. 6. Students can respond to questions or polls in real time through their own devices, using a provided platform or accessing a web-based application. It also offered diverse features like question types and polls, typically displaying results using charts, lists, etc. (Wong & Yunus, 2020). Classroom polling enables teachers to promptly pose questions and receive instant responses from students, encouraging participation from those who are hesitant to speak up, individuals with minority perspectives, and students who fear giving incorrect answers in front of their peers (Compton & Allen, 2018). These technologies require distinct interfaces to cater to the needs of both school teachers and students. The teachers' interface has been exclusive to teachers, allowing them to create questions/pools and view results ("Interactive presentation software," 2023). On the other hand, the students' interface is specifically designed for students to respond to questions posed by their teachers (Nearpod, 2023).

A vital component is the presentation screen, which displays questions and results to class students. The teachers have control of this screen, which is visible to teachers and students when connected to a projector (everywhere, 2023). A few also offered an additional screen exclusively for the teacher, enabling them to view live results before presenting them to the students and adjust presentation controls accordingly ("Interactive presentation software," 2023). Considering the above recommendations, various types of non-gamified RTs are available, such as Poll Everywhere (everywhere, 2023), Mentimeter ("Interactive presentation software," 2023), and Nearpod (Nearpod, 2023). We also compared the functionalities of these three popular tools (Compton & Allen, 2018; "Interactive presentation software," 2023; Nearpod, 2023; everywhere, 2023), as shown

Table 5 The functionality-based comparison between popular non-gamified RTs (referred to (Compton & Allen, 2018; “Interactive presentation software,” 2023; Nearpod, 2023; everywhere, 2023))

	Poll everywhere	Mentimeter	Nearpod
Pros	1. It also provides students with options to participate via text messages and offers various ways to analyze quantitative data 2. The teacher determines when students can see questions and sets time limits for poll completion	1. It offers a user-friendly interface with unlimited question types and respondents in the free account 2. PowerPoint slides can be imported as images. It supports image and video uploads, various question types, and a competitive quiz feature	1. It offers various interactive lesson options, including multimedia presentations, 3D objects, and quizzes 2. It also offers a library of pre-made lessons and resources created by teachers, providing a wealth of content that can be used for professional development and instructional support
Usefulness	1. It added flexibility integration with PowerPoint and Google Slides. However, compatibility depends on institutional policies and software versions 2. Teachers can export poll entries to an Excel file. The system's attractive design and diverse poll types make it appealing	1. It stands out as the most user-friendly tool, offering multiple question formats that include moderation, profanity filters, and options for teachers 2. This tool is highly effective when integrated into an online presentation platform that provides excellent instructional materials and support for teachers	1. It is a handy cloud-based platform that allows teachers to create engaging and interactive lessons and incorporate multimedia content 2. Teachers can create dynamic and personalized learning experiences, track student progress, and foster a more interactive and immersive classroom environment
Cost (Various subscription options)	Intro: Free, create unlimited questions, 25 maximum audience size Engage (in addition to intro): \$42 per month, 700 maximum audience size, create reports, access archived poll responses, and restrict responses to registered participants Teams (in addition to engage): \$84 per month, choose your response URL, Pinned questions during presentations	Intro: free, 50 participants/month, word clouds, polls, quizzes, and slides Basic (in addition to intro): \$11.99 per month, unlimited participants/month, import slides from other tools, and export results to Excel Pro (in addition to basic): \$24.99 monthly, co-edit presentations with your team members, and add your company logo and branding	Intro: free, 100 MB storage, 40 students maximum, creation of interactive slides, interactive videos, premade slides-based lessons, three teaching modes to support in-person, hybrid, and remote learning Gold (in addition to intro): \$13.25 per month, 1 GB storage, 75 students Platinum (in addition to intro): \$20.99 per month, 5 GB storage, 90 student maximum
Responses	1. The diverse range of poll activities helps to sustain student engagement 2. Rating activities prioritize students' interests and gauge the overall mood 3. Icebreaker activities facilitate smooth transitions to group-based tasks and enhance student confidence in using the tool	1. The 'Reaction' feature allows testing classroom engagement by gauging agreement or disagreement with displayed statements 2. Asynchronous open-question activities provide insights into student views and serve as a research tool for data comparison and cross-referencing	1. It offers a range of interactive features for student engagement, including multiple-choice and open-ended questions, polls, drawing tools, fill-in-the-blank exercises, matching pairs, collaborative boards, word clouds, hotspot images, and memory tests
Cons	1. The free version lacks moderation features and restricts responses depending on the plan 2. There is no limit to the number of polls that can be created. Adding questions during a presentation may disrupt the flow 3. Multimedia options are limited	1. Mentimeter's PowerPoint integration restricts teachers to a few Mentimeter questions 2. Free accounts have limitations regarding poll activities and quiz questions 3. Custom branding and colors require a paid subscription	1. It offers various interactive features, and the level of customization for activities and presentations may be limited compared to other platforms 2. A few features of Nearpod come at a price, and some schools or teachers may find the subscription fees costly, especially for larger-scale implementation

in Table 5. Non-gamified RTs foster learning by engaging school students and stepping away from passive listening. Unlike traditional methods, these technologies aid students' participation despite their place in the classroom, eliminating the need for hand-raising and public speaking (Chou et al., 2017; Dewi et al., 2023). Nevertheless, when these technologies are used repetitively, school students may lose interest. The repetitive nature of answering multiple-choice questions and the lack of variation leads to reduced engagement and learning motivation (Compton & Allen, 2018). Secondly, these technologies advance bias or social pressure, where students feel compelled to correspond to prevalent choices or sentiments rather than express their own reflections or perceptions, which can hinder engagement and critical thinking (Turan & Meral, 2018).

Gamified response technologies

The frequent utilization of non-gamified RTs in school classrooms may lead to decreased student interest (Cakiroglu et al., 2018; Janković & Lambić, 2022). The gamified classrooms enhance student engagement through enjoyable activities, incorporating game mechanisms (Barrio et al., 2015; Janković et al., 2023; Rahardja et al., 2019). Different models regarding gaming elements, i.e., a dynamical model for gamification of learning (Kim & Lee, 2015), 7GOALS (Gamification-oriented active learning steps; Oliveira et al., 2021), conceptual model (Zaric et al., 2017), engagement framework for reflective learning (Silpasuwanchai et al., 2016), game design framework (Huang & Hew, 2018), gamification output elements (Nurtanto et al., 2021), mechanics, dynamics, and emotions

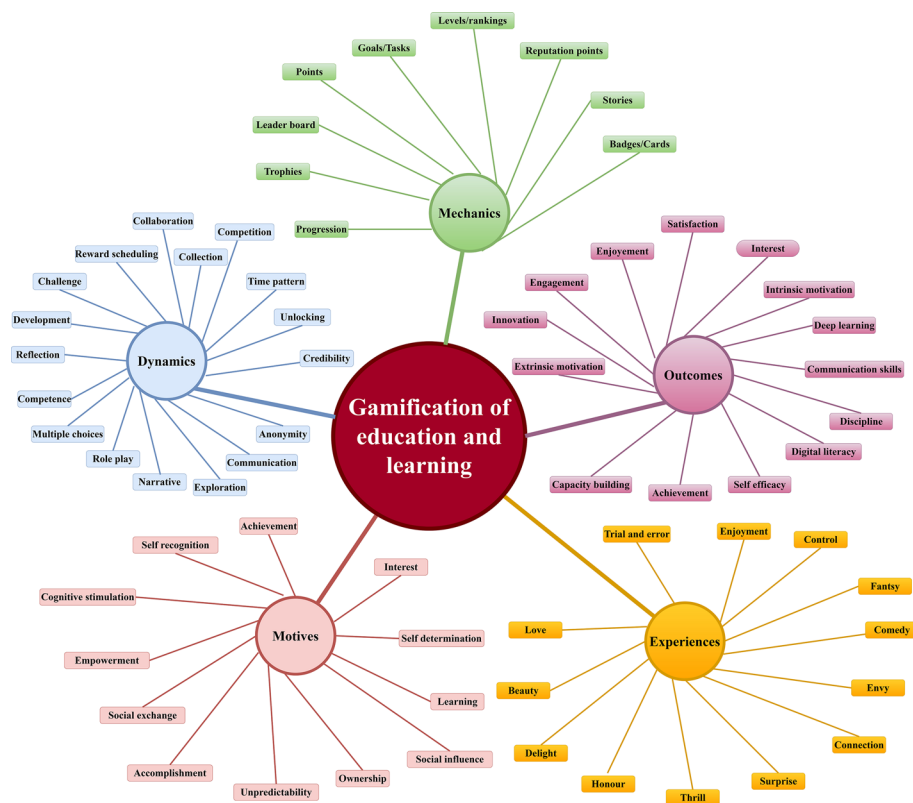


Fig. 7 The taxonomy of gamification elements for learning in education

(MDE) framework (Robson et al., 2015) were proposed in previous research. The common elements related to gamification for learning in school classrooms are presented in Fig. 7. New RTs are being developed to enhance student's learning capacity, including gamified RTs, which have gained popularity with the widespread use of mobile devices (Katemba & Sinuhaji, 2021; ÖDEN et al., 2021). These RT applications offer interactive experiences in the classroom, incorporating game-like structures to increase student engagement and motivation (Siwalete, 2021; Umboh et al., 2021). The level of student engagement substantially influences their achievement and performance in assessments (Malik et al., 2021).

Kahoot is a powerful gamified tool in education within and outside the classroom. It permits teachers to establish interactive question–answer games that require students to respond within a time limit (Curto Prieto et al., 2019). With its attractive interface, Kahoot has acquired pull as a classroom resource. Students take part utilizing electronic devices with internet access. (Setiawan & Soeharto, 2020) Recent research highlighted the advantages of Kahoot, such as fun, enjoyment, and excitement (Curto Prieto et al., 2019). Quizizz, another gamified RT, enabled numerous participants to join the platform simultaneously with an interactive and entertaining interface. By the end of 2020, Quizizz had crossed over 65 million monthly functioning users beyond 150 countries. It offers statistical data on quiz results and tractability regarding time for administering quizzes (Quizizz, 2023).

Quizalize is another educational tool that proposes gamified quizzes for students in the classroom. It provides school teachers with a user-friendly interface to generate, customize, and delegate quizzes to their students (España-Delgado, 2023). As each tool has different functionalities, we compared them based on various parameters (as mentioned in Kahoot, 2023; Quizalize, 2023; Quizizz, 2023)) as shown in Table 6. There are a few limitations to consider when using Gamified RTs. Some students may not be motivated by game elements and may not actively participate (Malik et al., 2021; Rahmawati et al., 2020). It may also lead to a focus on extrinsic rewards rather than intrinsic motivation for learning. The risk of overemphasizing competition can create anxiety or discourage collaboration (Sun & Hsieh, 2018). Furthermore, designing practical gamified activities requires careful planning and instructional design expertise (Jones et al., 2019; Lee et al., 2019). Researchers reported that some students, especially teachers, find it challenging to navigate different features of gamified RTs (Gustafsson & Ryve, 2022; Obery et al., 2021; Setiyani et al., 2020).

Effectiveness of RTs in school classrooms

Traditional lecture-based classrooms have limitations in sustaining students' attention and promoting student interaction (Chou et al., 2017; Shaw et al., 2019). Researchers highlighted that the integration of RTs in learning environments enhanced learning outcomes (Harefa et al., 2020; Ningsih & Mulyono, 2019). In the following sections, we will explain the effectiveness (significance and non-significance) of RTs at the primary, middle, and high school levels.

Table 6 The functionality-based comparison among gamified RTs (referred to Kahoot, 2023; Quizalize, 2023; Quizizz, 2023))

	Kahoot	Quizizz	Quizalize
Requires administrator account	Yes	No	Yes
Mcq's	Yes	Yes	Yes
Mcq's (Multi-selection)	Yes	Yes	Yes
Short text response	Only premium version	No	No
Fill in the blank	No	Yes	No
Open text responses	Only premium version (limited text)	limited text	No
Image-based questions	Yes	Yes	Yes
Video-based questions	Yes	No	No
Sorting	Only premium version	No	No
Voting	Only premium version	No	No
Feedback	Right or wrong	Partial	Partial
Open text questions evaluation	No	Partial	No
File export options	• Spreadsheet • Google Drive	• Print out • Download	• Export • Print • Result sharing (all paid)
File importing options	• Spreadsheets • Slides	• Spreadsheets • The teleport feature	• Spreadsheets • From Quizlet
Questions length	Limited	Unlimited	Limited
Questions timer	5–120 S	5 s–15 min	Dynamic
Scoreboard	Shown after each question	It is shown after each question but can be fixed	No
Team mode	Yes	No	No
Questions were viewed on the student's device	No	Yes	Yes
Student Engagement	Encourages active participation and competition.	Encourages active participation and self-paced learning.	Encourages active participation and self-paced learning.
Cost	\$7.99 per month	Customized	\$2.49 per month
AI-enabled	Yes	No	No

Primary school classrooms

Researchers suggested that digital tools have a significant role in students' learning to achieve their educational goals. Still, there is variation in digital tool usage among different school levels (Ningsih & Mulyono, 2019). In this context, RTs are pivotal in creating interactive and engaging learning experiences for student's (Djannah, 2021). Various articles indicated that primary school students respond enthusiastically to interactive learning activities facilitated by RTs (Abdullah et al., 2022; Dociu & Mihut-Kert, 2021; Pratama, 2021). The successful implementation of RTs depends upon the teacher's attitude towards technology (Agbatogun, 2014b), school culture, technological support and resources (Janković & Lambić, 2022; Ningsih & Mulyono, 2019). (Cakiroglu et al., 2018) reported that hand-held clickers had non-significant impact on participation rates, with only a few students showed higher or lower levels of participation. Similarly, (Sheard & Chambers, 2014) highlighted that hand-held clickers had no significant effects on high achievers. Students may not be able to learn through RTs if fewer participants

are involved (Wong & Yunus, 2020; Md Yunus & Azmanuddin bin Azman, 2019) that leads to a lack of peer interaction. A pilot study in Mathematics and English classes found that the non-gamified RT improved student engagement (Lu et al., 2019). Conversely, another study reported that although student participation and involvement were increased, it did not lead to better grades (Stanojević & Randelović, 2018). Teachers and students received immediate feedback on individual understanding of grammatical concepts using RT, but teachers' competence, classroom settings, and pedagogical technique are also crucial reported by teachers (Agbatogun, 2014b; Janković & Lambić, 2022). Researchers reported that knowledge retention is crucial in RTs and hasn't been extensively studied in this context (Md Yunus & Azmanuddin bin Azman, 2019). Learning retention refers to storing information in long-term memory for future recall. If not retained, data remains in short-term (working memory) memory and is quickly forgotten. Research shows that primary school students fail a significant portion of what they learn within a short period (González-Betancor & López-Puig, 2016). The cone of learning diagram developed by the National Training Laboratory demonstrated that specific teaching methods are more effective for long-term retention (Lalley & Miller, 2007). The retention of information can be improved by combining various elements such as badges and rewards, audio-visual learning, demonstrations, and practice, as shown in Fig. 8.

Integrating gamification features into learning techniques increased creative thinking skills (Sartono et al., 2021) and learning motivation for Arabic vocabulary (Fiani et al., 2021). Furthermore, speed and competition distract the students from in-depth understanding, which leads to insignificant results and anxiety among students. Additionally, the game-based format encouraged guessing rather than accurately reflecting a student's understanding of the material (Md Yunus & Azmanuddin bin Azman, 2019). When compared, the effectiveness of two gamified RTs (Kahoot and Quizizz) Kahoot group has a better conceptual understanding of the content with slightly higher scores than other group (Janković et al., 2023). Conversely, (Wulandari & Hartono, 2021) reported no difference in learning outcomes between students that used two different gamified RTs. (Janković & Lambić, 2022) pointed out no significant differences in academic achievement between the gamified RT and other groups. Similarly, there was no statistically significant difference observed between the lesson with non-gamified versus traditional tools and even students expressed that they had learned more with traditional tools (Měkota & Marada, 2020; Md Yunus & Azmanuddin bin Azman, 2019). Trends

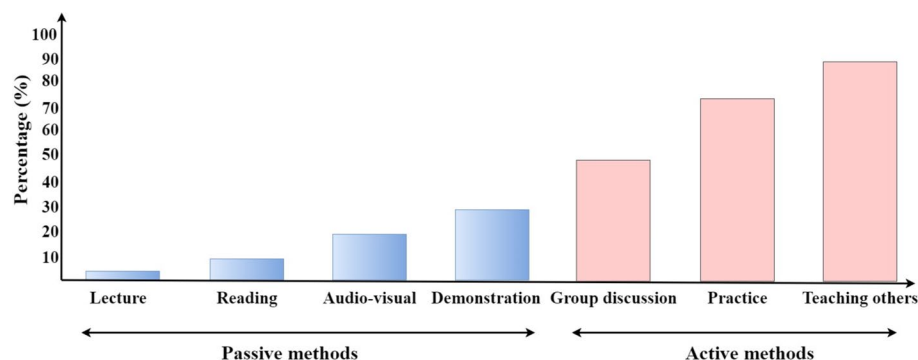


Fig. 8 Student learning taxonomy

showed that the successful integration of RTs largely depends on student involvement, lecture content, lecture delivery and teacher competence (Agbatogun, 2014b; Janković & Lambić, 2022). (Cahyaningtyas & Chakim, 2023) found no statistically significant difference between male and female students' perceptions concerning gamified RT. However, another study reported that girls improved more in learning irregular past verbs via gamified RT than boys, with a higher mean post-test score (Anak Yunus & Hua, 2021).

RT interventions were commonly employed in upper primary grades (grades 3–5), where students may benefit from more interactive and engaging instructional approaches. RTs often reinforce key concepts in these grades, assess student understanding, and promote active participation in classroom discussions (Agbatogun, 2014a). However, the effectiveness of Gamified RTs varied when compared to each other, leading to the fact that the choice of tool would be critical for effectiveness (Janković & Lambić, 2022; Janković et al., 2023). It has been observed that fewer articles are employed in lower primary grades (grades 1–2) due to concerns about students' developmental readiness and technical proficiency (Feldman, 2004). Various studies pointed out that gamified RTs are critical for students' assessment because immediate feedback and fast-paced interaction result in insignificant results, and complex, higher-order thinking skills are critical for evaluation (Feldman, 2004; Janković et al., 2023). We reported the summary of relevant literature that utilized RTs in primary school classrooms in Table 7. Various studies reported that a notable concern is the digital divide, which refers to disparities in access to RTs among students from different socioeconomic backgrounds (Chou, 2017; Setiawan & Soeharto, 2020; Md Yunus & Azmanuddin bin Azman, 2019). Younger students need more technical proficiency and help to operate these technologies without interruption during lectures (Janković et al., 2023; Sinnivasagam & Hua, 2023) that might lead to potential errors (Feldman, 2004). An over-reliance on RTs may inhibit the development of other key skills, such as critical thinking, problem-solving, analytical reasoning, etc. (Ningsih & Mulyono, 2019). Researchers emphasized that more experiments have been required to establish the RT's long-term impact on student engagement and learning outcomes (Cakiroglu et al., 2018; Ghanaat Pisheh et al., 2019; Janković & Lambić, 2022; Stanojević & Randelović, 2018).

Middle school classrooms

Several studies reported the potential benefits of RTs that support teaching and learning, as well as certain disadvantages (less meaningful conversation among peers (Barth-Cohen et al., 2016), no effect on extrinsic motivation (Sun & Hsieh, 2018), no significant difference in terms of learning attitudes (Hu & Huang, 2015) etc.). The results from the post-activity questionnaire on enjoyment revealed moderate satisfaction and lectures can be enjoyable using this approach (Balaskas et al., 2023). Similarly, students expressed that integrating RTs should only be used for fun, not learning purposes (Videnovik et al., 2018). The most noticeable benefits were observed among lower-performing students, where hand-held clickers promoted greater engagement that revealed the clicker-assisted teaching method was particularly effective for lower-performing students (Ghanaat Pisheh et al., 2019). Teachers found the occasional use of RTs in mathematics courses engaging and beneficial for understanding specific topics, but regular use in each lecture can lead to disengagement (Zhu & Urhahne, 2018). (Xin & Johnson, 2015)

Table 7 Summary of articles utilizing RTs in primary school classrooms

Reference (Author(s), year)	Course	Grade	RTs type	Study design	Sample size	Section's count	Evaluation method			Evaluation metrics		
							■	▣	●	○	◎	⊙
(Sinnivasagam & Hua, 2023)	Language	NM	*	Cross-sectional	150	NM	✓		✓	✓		✓
(Harahap & Kembaren, 2023)	Language	NM	*	Case study	10	NM	✓	✓	✓	✓		✓
(Janković et al., 2023)	Science	III	*	Quasi-experimental	72	02	✓					✓
(Yanuarto et al., 2023)	Mathematics	NM	*	Quasi-experimental	141	01	✓		✓	✓		✓
(Cahyaningtyas & Chakim, 2023)	Language	III	*	Descriptive	66	01	✓		✓	✓		✓
(Watson & Berry, 2022)	Science	IV, V	◆	Experimental	04	01	✓					✓
(Abdullah et al., 2022)	Language	I, II, V	◆	Experimental	54	03		✓				✓
(Janković & Lambić, 2022)	Science	III	*	Quasi-experimental	83	02	✓					✓
(Mahmud & Law, 2022)	Mathematics	I to V	*	Descriptive	50	NM	✓		✓	✓		✓
(Sartono et al., 2021)	Social science	V	*	Quasi-experimental	55	02	✓		✓	✓		✓
(Dociu & Mihut-Kert, 2021)	Language	V	*	Experimental	25	01	✓		✓	✓		✓
(Fiani et al., 2021)	Language	V	*	Descriptive	25	01	✓					✓
(Pratama, 2021)	Social studies	IV	*	Action	23	01	✓		✓	✓		✓
(Siwalete, 2021)	History	V	*	Action	20	01	✓		✓	✓		✓
(Wulandari & Hartono, 2021)	Mathematics	IV	*	Quasi-experimental	51	02	✓		✓	✓		✓
(Anak Yunus & Hua, 2021)	Language	V	*	Quasi-experimental	30	01	✓		✓	✓		✓
(Djannah, 2021)	Science	V	*	Developmental	149	05	✓		✓	✓		✓
(Umboh et al., 2021)	Mathematics	IV	*	Action	22	02	✓		✓	✓		✓
(Halim et al., 2020)	Language	NM	*	Quasi-experimental	60	02	✓		✓	✓		✓
(Idris et al., 2020)	Language	III	*	Experimental	31	01	✓		✓	✓		✓
(Setiawan & Soeharto, 2020)	Mathematics	IV	*	Action	28	01	✓		✓	✓		✓
(Mekota & Marada, 2020)	Geography	I to V	◆	Comparative	45	02	✓					✓
(Sukowati & Sartono, 2020)	Science	V	*	Action	22	01	✓		✓	✓		✓

Table 7 (continued)

Reference (Author(s), year)	Course	Grade	RTs type	Study design	Sample size	Section's count	Evaluation method			Evaluation metrics		
							■	▣	▢	●	○	◎
(Wong & Yunus, 2020)	Language	V	◆	Action	40	01	✓	✓	✓	✓	✓	✓
(Lu et al., 2019)	Mathematics Language	II	◆	Pilot	04	02	✓			✓	✓	
(Ningsih & Mulyono, 2019)	Language	NM	★	Exploratory	18	NM		✓				✓
(Hakim et al., 2019)	Social studies	IV	★	Descriptive	32	01	✓	✓		✓		✓
(Md Yunus & Azmanuddin bin Azman, 2019)	Language	III	★	Quasi-experimental	35	01	✓	✓		✓	✓	
(Stanojević & Randelović, 2018)	Science	III	◆	Quasi-experimental	26	01	✓			✓	✓	
(Cakiroglu et al., 2018)	Language	V	◆	Quasi-experimental	20	02	✓			✓	✓	
(Hou, 2018)	Language	III, IV, V	★	Quasi-experimental	42	01	✓			✓	✓	
(Chou, 2017)	Science	IV	◆	Experimental	20	01	✓		✓	✓	✓	✓
(Agbatogun, 2014a)	Language	NM	◆	Quasi-experimental	41	01	✓			✓	✓	
(Agbatogun, 2014b)	Language	NM	◆	Descriptive	17	01	✓		✓	✓	✓	
(Sheard & Chambers, 2014)	Language	V	◆	Pilot	187	11	✓	✓	✓	✓	✓	✓

NM = Not mentioned; RTs type: Hand-held clickers = ◆, non-gamified RTs = ◆, gamified RTs = ★

Evaluation method: Quantitative = ■, Qualitative = ▣

Evaluation metrics: Descriptive = ▢, Inferential = ○, Thematic = ◎

highlighted the positive outcomes of using hand-held clickers to engage middle school students with hyperactivity and behavioural disorders. However, this was not translated into measurable gains in students' academic achievement. Additional strategies, such as individualized feedback or cognitive interventions (Zhu & Urhahne, 2018), may be required to improve academic outcomes alongside RTs (Nordin, 2023). The immediacy of feedback provided by RTs allows teachers to adjust the lecture in real-time, addressing misconceptions and tailoring learning experiences to student needs (Koç & Sütçü, 2023).

A quasi-experimental study revealed significant differences in cognitive learning outcomes between control and RT groups (Rosyida et al., 2022). Statistical analysis showed a significant gender difference in students' vocabulary enhancement through gamified RT, but the male group tended to be noisier (Katemba et al., 2022). Researchers reported increased student engagement and promoted a formative assessment perspective (Obery et al., 2021) but results were not positive in the case of summative assessment (P.-N. Chou et al., 2017). Furthermore, (Da-Oh et al., 2023) investigated students' grammatical knowledge using Quizizz, and the results exhibited significant improvement in English grammar knowledge. More results can be improved by employing diverse instructional strategies and tools to enhance the learning experience and foster learning (Balaskas et al., 2023; Kartini & Sriyanto, 2023). Similarly, teachers reported that evaluation results of the assessment after RT intervention indicated that students' conceptual understanding remains suboptimal, as they were still passive in engaging with the learning process (Wulandari et al., 2023). The analysis showed no significant differences between the effectiveness of different gamified RTs (Obery et al., 2021) but significant difference between math and science courses exists (Curto Prieto et al., 2019). An overview of articles examining the use of RTs in middle school classrooms is presented in Table 8.

Researchers reported non-significant results after the RT intervention due to factors such as the digital divide, technical challenges, infrastructural challenges, etc., [e.g. (Hu & Huang, 2015; Sun & Hsieh, 2018; Videnovik et al., 2018)]. Moreover, attention should be paid to scaffolding students' digital literacy skills to ensure they can effectively navigate RT platforms and resources (Cohn & Fraser, 2016). However, teachers must be mindful of balancing RT usage with other instructional methods to prevent overreliance on technology (Chou et al., 2017; Tsihouridis et al., 2018). Additionally, targeted support may be necessary to address any technology-related barriers or challenges students encounter (Baguio et al., 2021). Furthermore, to better ascertain the effectiveness of RT-oriented teaching, verbal lessons should be prepared beforehand by teachers (Baguio et al., 2021). (Gustafsson & Ryve, 2022) developed design principles for questions for interactive lectures. The appropriate questions might be helpful for productive classroom practices according to different class knowledge levels. The limitations of the RTs affected student responses, as the small screens and limited character input hindered open-text responses (Chou et al., 2017; Xin & Johnson, 2015). Tracking student progress and correlating experiences with learning performance is challenging due to anonymous data collection and different grade-level participation across classrooms, preventing linking responses over time (Chou et al., 2017).

Table 8 A review of articles that explored the applications of RTs in middle school classrooms

Reference (Author(s), year)	Course	Grade	RTs type	Study design	Sample size	Section's count	Evaluation method			Evaluation metrics		
							■	▨	□	○	●	●
(Balaskas et al., 2023)	Science	VI	*	Experimental	27	01	✓			✓		
(España-Delgado, 2023)	Language	VI	*	Mixed-methods	27	01	✓	✓		✓		✓
(Kartini & Sriyanto, 2023)	Social studies	VII	*	Experimental	34	01	✓			✓		
(Nordin, 2023)	Language	NM	*	Quasi-experimental	33	01	✓			✓		
(Da-Oh et al., 2023)	Language	VIII	*	Quasi-experimental	35	01	✓			✓		
(Wulandari et al., 2023)	Mathematics	VIII	*	Developmental	12	01	✓			✓		
(Koç & Sütçü, 2023)	Language	VI	*	Quasi-experimental	40	01	✓	✓		✓		✓
(Sun & Lin, 2022)	Science	VII	◆	Experimental	58	02	✓			✓		
(Rusmardiana et al., 2022)	Social studies	VIII	*	Descriptive	68	02	✓			✓		✓
(Gustafsson & Ryve, 2022)	Mathematics	VIII	◆	Pilot	29	01	✓	✓		✓		
(Rosyida et al., 2022)	Science	VIII	*	Quasi-experimental	32	01	✓			✓		
(Sénécal et al., 2022)	Language	VIII	◆	Experimental	31	01	✓			✓		
(Pérez-Segura et al., 2022)	Language	VI	◆	Quasi-experimental	68	01	✓			✓		
(Katamba et al., 2022)	Language	VII	*	Quasi-experimental	69	02	✓			✓		
(Malayas et al., 2022)	NM	VII	◆	Cross-sectional	31	01	✓			✓		
(Shyr et al., 2021)	Mathematics	VII	*	Mixed-methods	24	02	✓	✓		✓		✓
(Rochimah & Muslim, 2021)	English	VII	*	Experimental	05	01	✓	✓		✓		✓
(Baguio et al., 2021)	Mathematics	VIII	*	Experimental	33	01	✓	✓		✓		
(Obery et al., 2021)	Science	VI, VII, VIII	*	Mixed-methods	42	02	✓	✓		✓		✓
(Zainuddin et al., 2020)	Science	VIII	*	Quasi-experimental	94	03	✓	✓		✓		✓
(Mays et al., 2020)	Language	VI	*	Quasi-experimental	24	01	✓	✓		✓		
(Curto Prieto et al., 2019)	Science	VI, VII, VIII	*	Experimental	68	02	✓			✓		
	Mathematics						✓			✓		

Table 8 (continued)

Reference (Author(s), year)	Course	Grade	RTs type	Study design	Sample size	Section's count	Evaluation method			Evaluation metrics		
							■	▣	▢	●	○	●
(Ghanaat-Pisheh et al., 2019)	Science	VIII	◆	Quasi-experimental	156	03	✓				✓	
(Zhu & Urhahne, 2018)	Mathematics	VI	◆	Quasi-experimental	170	07	✓			✓	✓	
(Videnovik et al., 2018)	NM	NM	★	Experimental	89	02	✓			✓	✓	
(Sun & Hsieh, 2018)	Language	VII	★	Quasi-experimental	75	02	✓		✓	✓	✓	✓
(Turan & Meral, 2018)	Social studies	VII	★	Quasi-experimental	46	01	✓			✓	✓	
(Tsihouridis et al., 2018)	Science	NM	★	Quasi-experimental	41	01	✓			✓	✓	
(Chou et al., 2017)	Language	VIII	◆	Quasi-experimental	24	01	✓		✓	✓	✓	✓
(Cohn & Fraser, 2016)	Science	VI, VIII	◆	Quasi-experimental	532	24	✓			✓	✓	
(Barth-Cohen et al., 2016)	Science	NM	◆	Experimental	137	08	✓		✓	✓	✓	✓
(Xin & Johnson, 2015)	Mathematics Language	VIII	◆	Experimental	05	02	✓		✓	✓	✓	✓
(Hu & Huang, 2015)	Science	VIII	◆	Experimental	43	01	✓			✓	✓	

NM = Not mentioned; RTs type: Hand-held clickers = ◆, non-gamified RTs = ◆, gamified RTs = ★

Evaluation method: Quantitative = ■, Qualitative = ▣

Evaluation metrics: Descriptive = ▢, Inferential = ○, Thematic = ●

High school classrooms

According to a National Center for Education Statistics survey, many high school teachers in the United States used educational technologies in their classrooms during the 2019–2020 academic year (Gray & Lewis, 2021). Gamified RT (Kahoot) in biology classes positively impacted student engagement and learning, making complex concepts more enjoyable (Jones et al., 2019). However, Mathematics students indicated that hand-held clickers did not positively impact their ability to respond when learning new course content and nor improve the completion rate of classwork (Wang et al., 2014). Moreover, incorporating RTs did not resolve all the problems that hindered the learning process. To enhance students' meta-cognitive understanding of sciences related concepts, questions should encompass observable phenomena and the underlying principles that can be reasoned and explained (Binek et al., 2016). Furthermore, (Nadhifah & Jauhariyah, 2021) recommended that the assessment through Quizizz is feasible, particularly in explaining scientific phenomena and designing and evaluating scientific investigations. Hand-held clickers provide immediate feedback and encourage student participation and engagement in physics and Geometry classes (Wang et al., 2014). A study found a non-significant correlation between summative test grades and students' positive perception of gamified RT that suggested students who recommended the intervention didn't achieved high grades (Catindig & Prudente, 2019). Similarly, statistical analysis clearly showed that the use of gamified RT resulted in no significant change in the student's scores (Anh et al., 2021).

Researchers explored the implementation of Kahoot in English vocabulary learning (Quiroz et al., 2021), which found positive results in language learning among high school students, indicated potential benefits in improving reading comprehension, and addressed the weaknesses in language skills (Mansur & Fadhilawati, 2019). A study claimed that the activities carried out with gamified RT have no significant effect on the student's language learning motivation and also do not have a significant impact on exam anxiety (ÖDEN et al., 2021). Furthermore, (Buele-Guerrero & Torres-Carrión, 2022) argued that a question asked through RT during the lectures with a high rate of correct responses didn't guarantee full understanding of the concept. Therefore, it's recommended that teachers review each question asked through RTs immediately after displaying the correct answer (Maming et al., 2023). Students stated that the use of Quizizz for understanding energy and momentum concepts was not significantly different from traditional lectures (Yana et al., 2022). However, the lab and theory course results differed as it was argued that the course content and assessment design differed (Naumoska et al., 2022). Quantitative findings revealed that student's anxiety was not decreased significantly in the experiment group compared to the control group during English lectures (Can et al., 2021) but did not consistent with other studies (Turan & Meral, 2018). (Lee et al., 2019) conducted a study in a high school where experimental and control groups demonstrated similar performance in homework and exams after gamified RT was implemented, indicating that its integration did not enhance students' overall learning outcomes.

Statistical analysis did not reveal a significant difference in student engagement between the gamified and non-gamified RTs. Regarding gender, male students showed higher levels of motivation, attention, engagement, and learning performance when

using RTs compared to female students. This disparity might have been attributed to the fact that women were often less influenced by competitive motivational factors (Barrio et al., 2015). Researchers also reported that the results were partially positive or neutral after RT intervention as participants perceived that learning in a new way is difficult and sometimes attention being diverted from the lecture (Can et al., 2021; Mansur & Fadhilawati, 2019). Students were concerned that the integration of such type of interaction limits the interaction between teachers and students, unable to review quiz questions for correction and time constraints for each question (Al Munawaroh, 2021). To promptly enhance students' grasp of essential concepts, it was recommended to structure the questions in RTs as a series and repeat the cycle various times within a lesson (Binek et al., 2016). The intervention of RTs as a formative assessment tool throughout the lesson aids teachers in assessing students' learning progress more effectively (Buele-Guerrero & Torres-Carrión, 2022; Hikmah et al., 2021; Suryani & Fauziati, 2022). However, (Chiang, 2020) reported that students felt that RTs should not be used as assessment tools as they allow participation even when not physically present, and students became overly enthusiastic and shouted out the answers.

Table 9 presents a summary of articles investigating the utilization of RTs in high school classrooms. The challenges faced by high school teachers should be addressed through redesigned professional development programs that prioritize accessibility, interactivity, and theory-informed design principles, providing enduring guidelines for teachers to enhance learning and engagement with RTs (Chiang, 2020; Naumoska et al., 2022; Suryani & Fauziati, 2022). Real-time response visualization for students is also crucial, especially when students struggle to find correct answers individually. There should be a balanced approach for displaying voting results when most students answer correctly and hiding them to encourage exploring diverse ideas when many students provide incorrect answers (Chien et al., 2015).

Discussion

General trends and divergences

Researchers reported that RTs have positive impacts on students motivation (Purba et al., 2019; Stoyanova et al., 2017), engagement, reduced anxiety (Turan & Meral, 2018), class participation (Malandrino et al., 2014), personalized feedback (Pérez-Segura et al., 2022), and learning experience (Sun & Hsieh, 2018), anonymous interaction, overcoming communication detention (Plump & LaRosa, 2017), attention (Sun & Hsieh, 2018), active learning (Malandrino et al., 2014), peer discussions (Hu & Huang, 2015) formative and summative assessment (Catindig & Prudente, 2019; Sheard & Chambers, 2014) and better than traditional learning environments (Al Munawaroh, 2021; dos Santos Teotônio et al., 2020). Conversely, few studies also reported no significant impact on students' engagement (Tarazi & Ortega-Martín, 2023), anxiety (Can et al., 2021), motivation (Sun & Hsieh, 2018), knowledge construction (Sun & Lin, 2022), learning achievement (Rahmahani, 2020), collaborative learning (Barth-Cohen et al., 2016), attention (Cahyani et al., 2022), active participation (Cakiroglu et al., 2018), conceptual understanding (Wulandari et al., 2023), and language learning (ÖDEN et al., 2021). Furthermore, researchers reported no significant differences between the effectiveness of different RTs when compared with each other (Barrio et al., 2015; Janković & Lambić, 2022; Obery

Table 9 A summary of articles that explored the application of RTs in high school classrooms

Reference (Author(s), year)	Course	Grade	RTs type	Study design	Sample size	Section's count	Evaluation method		Evaluation metrics		
							■	▣	●	○	◎
(Siswati et al., 2023)	Science	NM	◆	Quasi-experimental	27	01	✓		✓	✓	
(Tarazi & Ortega-Martín, 2023)	NM	NM	◆	Quasi-experimental	15	01	✓		✓	✓	
(Maming et al., 2023)	Language	IX	✱	Quasi-experimental	22	01	✓		✓	✓	
(Powa & Murniarti, 2022)	Mathematics	XI, XII	◆	Descriptive	144	04	✓		✓	✓	
(Buele-Guerrero & Torres-Carrión, 2022)	Science	X	✱	Descriptive	77	06	✓	✓	✓	✓	✓
(Naumoska et al., 2022)	Science	IX	◆	Mixed-methods	251	05	✓	✓	✓	✓	✓
(Cahyani et al., 2022)	Mathematics	XI	✱	Developmental	38	03	✓	✓	✓	✓	✓
(Suryani & Fauziati, 2022)	Language	IX	✱	Mixed-methods	30	02	✓	✓	✓	✓	✓
(Yana et al., 2022)	Science	X	✱	Descriptive	66	01	✓		✓	✓	
(ÖDEN et al., 2021)	Language	IX	✱	Quasi-experimental	44	01	✓	✓	✓	✓	✓
(Can et al., 2021)	Language	IX	◆	Mixed-methods	28	01	✓	✓	✓	✓	✓
(Akhsani & Yanuarto, 2021)	Mathematics	X	✱	Descriptive	NM	01	✓		✓	✓	
(Anh et al., 2021)	Science	X	✱	Case study	69	01	✓		✓	✓	
(Katemba & Sinuhaji, 2021)	Language	NM	✱	Experimental	32	01	✓		✓	✓	
(Hikmah et al., 2021)	Science	XI	✱	Developmental	33	01	✓		✓	✓	
(Quiroz et al., 2021)	Language	IX	✱	Quasi-experimental	29	01	✓		✓	✓	✓
(Al Munawaroh, 2021)	Language	X	✱	Descriptive	34	01	✓	✓	✓	✓	✓
(Malik et al., 2021)	Science	XI	✱	Developmental	53	02	✓		✓	✓	
(Dhamayanti, 2021)	Science	XI	✱	Descriptive	143	05	✓		✓	✓	
(Rahmahani, 2020)	Language	XII	✱	Experimental	127	05	✓		✓	✓	✓
(Chiang, 2020)	Language	XII	✱	Case study	65	01	✓	✓	✓	✓	✓

Table 9 (continued)

Reference (Author(s), year)	Course	Grade	RTs type	Study design	Sample size	Section's count	Evaluation method	Evaluation metrics
							■ ■ ■	● ○ ◎
(Rahmawati et al., 2020)	Mathematics	X	*	Developmental	NM	01	✓	✓
(Setiyani et al., 2020)	Mathematics	X	*	Quasi-experimental	32	01	✓	✓
(Harefa et al., 2020)	Science	NM	*	Quasi-experimental	38	01	✓	✓
(dos Santos Teotônio et al., 2020)	Mathematics	XI	*	Case study	30	01	✓	✓
(Catindig & Prudente, 2019)	Science	XII	*	Quasi-experimental	33	01	✓	✓
(Lee et al., 2019)	Science	IX	*	Case study	20	01	✓	✓
(Mansur & Fadhilawati, 2019)	Language	X	*	Action	40	01	✓	✓
(Purba et al., 2019)	Science	X	*	Quasi-experimental	40	01	✓	✓
(Jones et al., 2019)	Science	NM	*	Experimental	18	01	✓	✓
(Stoyanova et al., 2017)	Mathematics	XI	*	Experimental	78	01	✓	✓
(Binek et al., 2016)	Science	IX, X	◆	Experimental	621	24	✓	✓
(Barrio et al., 2015)	Science	NM	*	Experimental	55	01	✓✓	✓
(Chien et al., 2015)	Science	XI	◆	Quasi-experimental	87	02	✓	✓
(Agarwal et al., 2014)	Mathematics Language Science	IX, X, XI, XII	◆	Cross-sectional	102	NM	✓	✓
(Wang et al., 2014)	Mathematics	X, XI, XII	◆	Quasi-experimental	47	02	✓	✓
(Malandrino et al., 2014)	Mathematics	NM	◆	Experimental	17	01	✓	✓

NM = Not mentioned; RTs type: Hand-held clickers = ◆, non-gamified RTs = ◆, gamified RTs = *

Evaluation method: Quantitative = ■, Qualitative = ●

Evaluation metrics: Descriptive = ■, Inferential = ○, Thematic = ◎

et al., 2021; Wulandari & Hartono, 2021). Similarly, no statistically significant difference was observed between the lesson learned with RTs vs traditional methods (Hu & Huang, 2015; Měkota & Marada, 2020; Md Yunus & Azmanuddin bin Azman, 2019; Zainuddin et al., 2020; Stoyanova et al., 2017). Statistical analysis showed that students performed better with traditional learning methods rather than interactive lectures (Chou et al., 2017; dos Santos Teotônio et al., 2020). Several studies pointed out that there was no significant difference in students' perception based on gender for the effectiveness of RTs [e.g., (Chiang, 2020)], but (Katemba et al., 2022; Sinnivasagam & Hua, 2023) findings stated that gender is a significant factor, with boys being more interested to the use of RTs. Similarly, another study showed that women were often less influenced by competitive motivational factors (Barrio et al., 2015). Conversely, (Anak Yunus & Hua, 2021) reported that female students improved more in learning irregular past verbs than boys, with a higher mean post-test score after integration of RT. Researchers reported that RTs reduced anxiety (Turan & Meral, 2018) but these findings are not consistent or aligned with other reported results (España-Delgado, 2023; ÖDEN et al., 2021). These divergences in the reported results might be due to the high cost associated with RTs (Chien et al., 2015), low network or bandwidth (Shyr et al., 2021), lack of institutional support (Ningsih & Mulyono, 2019), limited digital competence (Měkota & Marada, 2020; Md Yunus & Azmanuddin bin Azman, 2019), or lack of teacher's experience (Agbatogun, 2014b; Cakiroglu et al., 2018).

It has been analyzed that most of the studies were conducted in high school classrooms then other school classrooms (primary and middle) in fewer articles. The fact that most of the researchers are high school teachers and can quickly gain the required authorization could be a significant factor in explaining the current scenario. High school students are more likely to own smartphones than other students, which could be another factor in the selection of this group (Chou et al., 2017; Statista, 2022). There are different numbers of participants in each article, which is evident that participation rates are low in particular studies but relatively high in others. Furthermore, it has been difficult to generalize the findings because different instructional practices were used. A large proportion of studies focused on peer instruction (e.g. Barth-Cohen et al., 2016; Chien et al., 2015; Shyr et al., 2021), gamification [e.g. (Halim et al., 2020; Idris et al., 2020)], but only a few studies focused on online (Mahmud & Law, 2022) or hybrid settings. It has also been discovered that most articles were in science courses especially in physics (Anh et al., 2021; Binek et al., 2016; Catindig & Prudente, 2019; Chien et al., 2015; Hikmah et al., 2021; Yana et al., 2022) and chemistry (Harefa et al., 2020; Naumoska et al., 2022; Purba et al., 2019; Rahmahani, 2020) subjects. In science courses, the level of questions differs from that of literature or language courses (Curto Prieto et al., 2019). Language education has been the second most popular courses, with English courses being its primary application. Most number of studies were conducted in English courses for vocabulary (Quiroz et al., 2021; Sénécal et al., 2022), grammar (Da-Oh et al., 2023; Koç & Sütçü, 2023; Sheard & Chambers, 2014), and comprehension (Mays et al., 2020; Rochimah & Muslim, 2021) learning. Other than English language, few studies integrated RTs in other languages courses such as German (Dociu & Mihut-Kert, 2021), Arabic (Fiani et al., 2021), and Indonesian (Cahyaningtyas & Chakim, 2023) learning. This might be the case because

RTs make providing engagement and learning motivation simple, two crucial components of language learning (Al Munawaroh, 2021). Additionally, there are RTs (e.g., Quizlet) designed specifically for language learning. Data synthesis showed that student learning motivation differed from school levels. Teachers were more focused on promoting active participation at primary school levels, but at middle or high school levels, questions are designed to evaluate students' critical thinking and reasoning skills (Quiroz et al., 2021).

Earlier researches were more focused on hand-held clickers [e.g., (Chou, 2017; Wang et al., 2014; Xin & Johnson, 2015) etc.,] but more studies have been conducted through software based-RTs in recent years [e.g., (Balaskas et al., 2023; Siswati et al., 2023) etc.,]. Among the types of RTs studied, gamified RTs were the most prevalent ($n=72$), then followed by non-gamified RTs ($n=18$), and the least utilized are hand-held clickers ($n=15$). Understanding this distribution provides insights into which RTs are commonly adopted, how they are implemented in school classrooms, and their impact on learning outcomes. Furthermore, (Plump & LaRosa, 2017) pointed out that the tool's initial availability as a free prescription may have significantly impacted the large number of articles that analyzed it. In addition to Kahoot, Mentimeter is another non-gamified RT that is often reviewed in articles. Like Kahoot, Mentimeter enhances learning motivation and performance (Tarazi & Ortega-Martín, 2023; Wong & Yunus, 2020). While most articles do not refer to Mentimeter as gamified RT (Turan & Meral, 2018), some claim it contains aspects of a game akin to Kahoot (Compton & Allen, 2018). Socrative's initial availability at no cost contributed to its inclusion in articles. Because preference may arise from being freely accessible to teachers and students (Suryani & Fauziati, 2022), unlike the other tools, Pool Everywhere, Nearpod, Turning Technologies, and iClicker, utilized in some of the reviewed articles, do not include entertainment elements (Wang & Tahir, 2020). Its low student engagement compared to other tools (Compton & Allen, 2018) could be why it receives less attention in the selected articles. The number of articles utilizing hand-held clickers in school classrooms is down due to high cost and emerging alternative technologies (Wang & Tahir, 2020). The selected articles are distributed over various years, providing insight into the temporal trends within the study. Figure 9 indicates the distribution of articles included in the study.

Studies type

Numerous articles used quantitative methods (such as the questionnaire or content-based assessment before and after intervention) to evaluate student motivation and engagement (e.g., Djannah, 2021; Siwalette, 2021). It was seen that the most popular way to assess effectiveness of RTs was through questionnaires from students using electronic survey platforms like Google Forms (Oates et al., 2022). Furthermore, the questionnaire might be the best tool for researching student perceptions regarding RTs (Harefa et al., 2020; Purba et al., 2019). Some articles included whole scales with over a dozen engagement-related items (Zainuddin et al., 2020); others only had a few related items (Catindig & Prudente, 2019). But, these articles more focused on "perceived" outcomes rather than actual (e.g., Balaskas et al., 2023; Sinnivasagam & Hua, 2023; Sun & Lin, 2022) etc.,). However, questionnaires are not necessarily the most accurate approach for evaluating student's perceptions (Akhsani & Yanuarto, 2021; Barrio et al.,

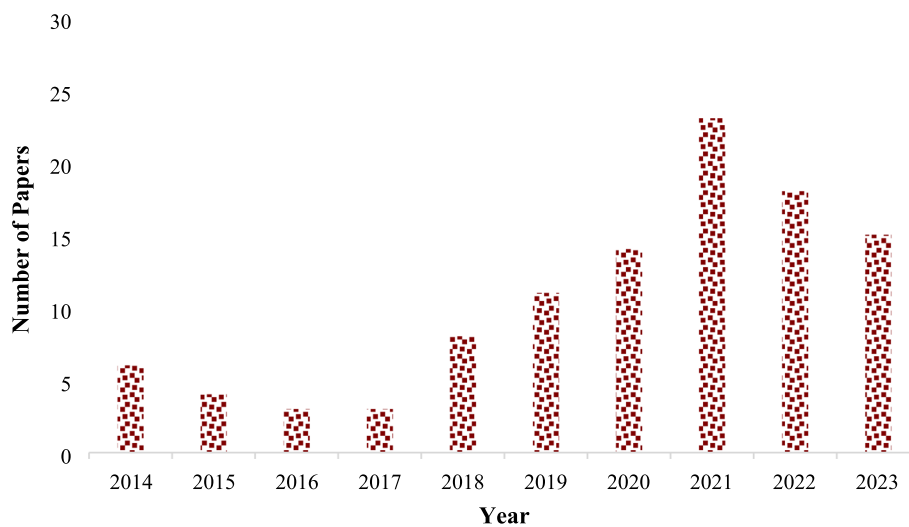


Fig. 9 The number of articles included by each year

2015). Researchers employed content-based assessment as evaluation method because they offer a robust method for assessing the impact of RTs and could identify the effectiveness before and after the intervention (e.g., Lu et al., 2019; Pérez-Segura et al., 2022; Sénécal et al., 2022)). It provides quantitative data that can directly measure the effect of an intervention (Oates et al., 2022). Qualitative measures were the second most common method of assessing student perceptions observed in the articles. Qualitative metrics are beneficial in exploratory research where it's difficult to identify or quantify student perceptions. Few articles employed qualitative measures to enable them to capture students and teachers' insights concerning benefits and challenges. However, one drawback of using qualitative methodologies is their difficulty in scale (Oates et al., 2022) but often leads to in depth thoughts of participants that could not possible through quantitative methods. Researchers are often limited in the amount of data due to certain challenges such as manual analysis of data, time-intensive, subjectivity in interpretation (Abdullah et al., 2022), limited generalizability, or influence of researcher presence during lectures (Ningsih & Mulyono, 2019). To overcome these issues and rigorous analysis, researchers deployed mixed-methods study designs that combined qualitative and quantitative procedures to provide a holistic research approach [e.g., (Obery et al., 2021; Shyr et al., 2021) etc.,]. According to (Oates et al., 2022), it helps to control potential confounding variables and allows researchers to establish causal relationships with excellent reliability and objectivity. However, researchers reported that it has also been challenging to align qualitative insights and quantitative metrics meaningfully (Can et al., 2021; Suryani & Fauziati, 2022).

Design of the studies

Different study designs have some positive and negative aspects but (Oates et al., 2022) argued the choice depends on study objectives, research questions, population characteristics, resources availability, ethical considerations, context and setting, and researcher's expertise. Researchers employed various study designs to investigate the

effectiveness of RTs in school classrooms at different levels. Data analysis showed that majority of the articles employed a quasi-experimental design that allowed researchers to compare control and experimental groups that were not randomly assigned (Oates et al., 2022) but provided crucial findings on the effectiveness of RTs (e.g., Rosyida et al., 2022; Zainuddin et al., 2020)). In addition, it also explains how RTs could be integrated into lectures to evaluate learning outcomes (Setiyani et al., 2020; Wang et al., 2014) but lacks random assignment and challenging to determine causality due to selection bias (Siswati et al., 2023; Wang et al., 2014). Several studies used experimental design in controlled educational settings to quantify the effectiveness of RTs due to higher internal validity and generalizability (e.g., Idris et al., 2020; Sénécal et al., 2022; Xin & Johnson, 2015)). Additionally, only a small number of studies have looked at how students perceived the RTs and how well it works with descriptive design through observational data (e.g., Al Munawaroh, 2021; Buele-Guerrero & Torres-Carrión, 2022; Hakim et al., 2019)). Furthermore, action research design was an iterative method implemented in few studies in which teachers assessed and refined various pedagogical practices based on student feedback and effective for improving educational practices directly as it combined practical implementation with researcher or teacher reflection to enhance students experiences with adaptive practices (e.g., (Pratama, 2021; Sukowati & Sartono, 2020; Mansur & Fadhilawati, 2019)). Researchers argued that case studies offered valuable insights into a particular learning settings but the effectiveness or outcomes concerning RTs are not generalizable to a larger educational environments (dos Santos Teotônio et al., 2020; Lee et al., 2019). It could be possible due to these concerns that only few studies employed case study design (e.g., dos Santos Teotônio et al., 2020; Harahap & Kembaren, 2023; Lee et al., 2019)). Developmental studies were primarily utilized to evaluate various aspects (e.g., usability, reliability, performance etc.,) of newly developed RTs to evaluate usability aspects through iterative testing and feedback but mostly employed within small classrooms with low number of students (e.g., Rahmawati et al., 2020; Wulandari et al., 2023)). Furthermore, a smaller number of pilot (Gustafsson & Ryve, 2022; Lu et al., 2019; Sheard & Chambers, 2014), cross-sectional (Agarwal et al., 2014; Malayas et al., 2022; Sinnivasagam & Hua, 2023), comparative (Mékota & Marada, 2020), and exploratory (Ningsih & Mulyono, 2019) studies were implemented in diverse levels of schools classrooms. The distribution of research designs in the study is detailed in Fig. 10, which shows the frequency of articles corresponding to each research design type, with quasi-experimental designs being the most prevalent ($n=38$), followed by experimental designs ($n=23$).

Prevalent themes

The themes were coded into engagement, learning motivation, learning performance, and challenges, as discussed in most articles. RTs are an educational tool that increases students' motivation, participation, and engagement in class by assisting them in reaching the target learning level (Malandrino et al., 2014; Plump & LaRosa, 2017; Sinnivasagam & Hua, 2023; Sun & Hsieh, 2018). Researchers reported that student engagement facilitates active participation and interaction, enhancing involvement in classroom activities (Catindig & Prudente, 2019; Malik et al., 2021). However,

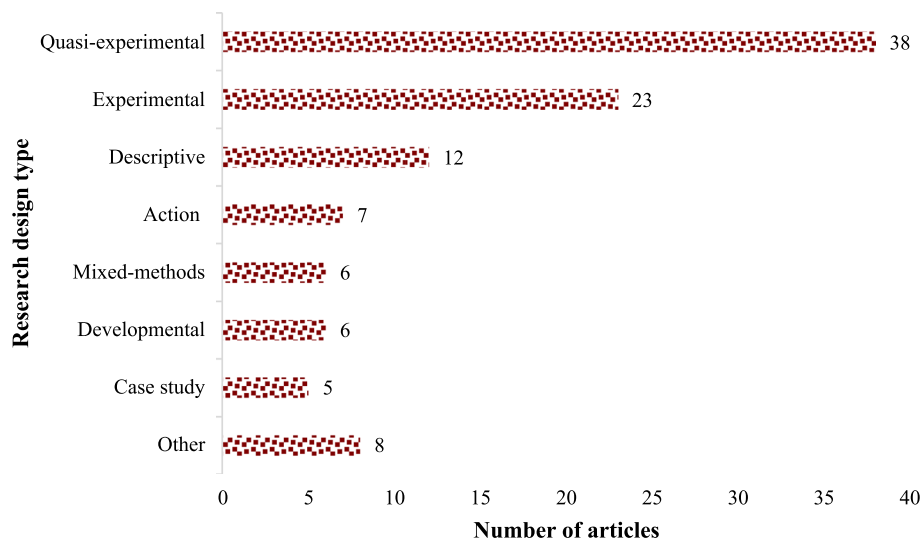


Fig. 10 Distribution of research design types in articles

concerns arise regarding the potential for RTs to foster surface-level engagement rather than deep learning (Curto Prieto et al., 2019; Jones et al., 2019; Mays et al., 2020; Sénécal et al., 2022). Learning performance is another aspect observed in the articles that yield important information about how well a course, exercise, or teaching resource is doing (Anak Yunus & Hua, 2021). Gamified RTs are an excellent way to improve student motivation through interactive features (Umboh et al., 2021). The findings suggested that RTs increase students' intrinsic motivation by providing immediate feedback (Naumoska et al., 2022) but not able to increase student's learning outcomes always (Sun & Lin, 2022). Nonetheless, the sustainability of this motivation remains to be determined over time (Sun & Lin, 2022). Content-based assessments are crucial for assessing student learning performance and the effectiveness of curricula and instructional strategies by demonstrating improvements in learning outcomes (Barth-Cohen et al., 2016).

Some studies have demonstrated positive effects on academic achievement (e.g., Baguio et al., 2021; Cohn & Fraser, 2016)), while others indicate no significant improvement based on grades (e.g., Al Munawaroh, 2021; Anh et al., 2021; Maming et al., 2023; Rahmahani, 2020)). Furthermore, the insignificant outcomes may be due to various challenges. Researchers reported financial challenges such as cost associated with RTs for institutions or lack of financial means to acquire a compatible device (Binek et al., 2016; Chien et al., 2015; Ningsih & Mulyono, 2019). The lack of proper infrastructure, e.g., network malfunctioning during lectures (Naumoska et al., 2022; Suryani & Fauziati, 2022) and lack of support staff (Janković & Lambić, 2022; Ningsih & Mulyono, 2019) may lead to insignificant learning outcomes. The findings of pilot studies reported that students and teachers encountered interface issues that hindered user interaction with RTs (Gustafsson & Ryve, 2022; Lu et al., 2019; Sheard & Chambers, 2014). Teachers reported various challenges like struggle to respond to a large volume of responses, especially using open-text questions for large classrooms (España-Delgado, 2023; Mahmud & Law, 2022; Sinnivasagam & Hua, 2023), less course content coverage due technological intervention (Harahap & Kembaren, 2023; Wulandari et al., 2023), time-intensive to design questions

(Gustafsson & Ryve, 2022; Nordin, 2023), and difficult to retake control of the class after the intervention (Agbatogun, 2014b). Students stated potential barriers i.e., learning in a new way was difficult (Can et al., 2021), reluctant to participate in discussions in interactive activities (Barrio et al., 2015; Mansur & Fadhilawati, 2019), distraction (Md Yunus & Azmanuddin bin Azman, 2019; Cahyani et al., 2022), real-time pressure to respond quickly (Can et al., 2021; España-Delgado, 2023), and peer pressure through visible results (Hu & Huang, 2015; Shyr et al., 2021). Overall, the findings suggested that it would be beneficial to continuously evaluate the RT's effectiveness during various time-spans in the semester to foster academic progress. Table 10 pertains to various themes, including engagement, learning motivation, learning performance, and challenges.

Limitations

A common limitation concerning SLRs is the potential for the research to be restricted due to the keywords used in the search query (i.e., Type I and Type II errors), the selection of bibliographic databases, and the time frame covered by the shortlisted studies. Additionally, only peer-reviewed papers were included for data synthesis, but it might be possible that gray or white literature can provide valuable insights. The process of synthesizing qualitative data coded, categorized, and analyzed may introduce subjectivity, potentially leading to differences in the overall interpretation of the effectiveness of RTs. However, other aspects could be further extracted and discussed from the shortlisted studies. Positive findings were more likely to be published than negative or inconclusive results, which could skew the overall assessment of its effectiveness.

Future guidelines for the implementation of RTs in school classrooms

The perceptions about RTs still differ despite attempts to integrate them into school classrooms to improve learning outcomes due to barriers like technological infrastructure, including non-functional computers or power outages, especially internet connectivity, etc. (Shyr et al., 2021; Xin & Johnson, 2015). The successful integration of RTs for students requires careful consideration from various stakeholders (technologists, teachers, institutions, and policymakers).

Recommendations for technologists

Technologists or developers are one of the stakeholders in designing RTs. Various studies highlighted issues concerning design (Feldman, 2004), certain functionalities (Lu et al., 2019), privacy and security protocols (Stanojević & Randelović, 2018), compatibility and adaptability (Sun & Hsieh, 2018; Zainuddin et al., 2020). The following recommendations could be valuable for technologists in designing and developing the RTs.

Human-centered design

Adults and children exhibit distinct cognitive abilities, particularly in attention and logical reasoning. Children's attention spans are shorter, requiring more interactive and engaging user interface designs (Lu et al., 2019). Due to children's less mature mental development, designers must ensure transparent user experiences concerning the following age groups, as suggested by (Feldman, 2004).

Table 10 Major themes extracted from selected articles

Themes	References
Engagement	Sinnivasagam and Hua, (2023), Harahap and Kembaren, (2023), Cahyaningtyas and Chakim, (2023), Watson and Berry, (2022), Mahmud and Law, (2022), Djannah, (2021), Měkota and Marada, (2020), Lu et al., (2019), Ningsih and Mulyono, (2019), Hakim et al., (2019), Balaskas et al., (2023), Sun and Lin, (2022), Sénécal et al., (2022), Malayas et al., (2022), Zainuddin et al., (2020), Mays et al., (2020), Curto Prieto et al., (2019), Zhu and Urhahne, (2018), Sun and Hsieh, (2018), Turan and Meral, (2018), Tarazi and Ortega-Martín, (2023), Powa and Murniarti, (2022), Yana et al., (2022), Malik et al., (2021), Dhamayanti, (2021), Rahmahani, (2020), Harefa et al., (2020), Jones et al., (2019), Malandrino et al., (2014)
Learning motivation	Sinnivasagam and Hua, (2023), Yanuarta et al., (2023), Abdullah et al., (2022), Sartono et al., (2021), Docu and Mihut-Kert, (2021), Fiani et al., (2021), Pratama, (2021), Siwalette, (2021), Halim et al., (2020), Idris et al., (2020), Setiawan and Soeharto, (2020), Sukowati & Sartono, (2020), Wong and Yunus, (2020), Hakim et al., (2019), Hou, (2018), Agbatogun, (2014b), Balaskas et al., (2023), Kartini and Sriyanto, (2023), Rusmardiana et al., (2022), Gustafsson and Ryve, (2022), Rosyida et al., (2022), Shyr et al., (2021), Pérez-Segura et al., (2022), Curto Prieto et al., (2019), Sun and Hsieh, (2018), Tsihouridis et al., (2018), Hu and Huang, (2015), Naumoska et al., (2022), ÖDEN et al., (2021), Akhsani and Yanuarta, (2021), Katemba and Sinuhaji, (2021), Dhamayanti, (2021), Rahmawati et al., (2020), Setiyani et al., (2020), dos Santos Teotônio et al., (2020), Mansur and Fadhilawati, (2019), Purba et al., (2019), Stoyanova et al., (2017), Binek et al., (2016), Barrio et al., (2015), Wang et al., (2014)
Learning performance	Harahap and Kembaren, (2023), Janković et al., (2023), Cahyaningtyas and Chakim, (2023), Janković and Lambić, (2022), Wulandari and Hartono, (2021), Anak Yunus and Hua, (2021), Umboh et al., (2021), Md Yunus & Azmanuddin bin Azman, (2019), Stanojević and Randelović, (2018), Chou, (2017), Agbatogun, (2014a), Sheard and Chambers, (2014), Kartini and Sriyanto, (2023), Nordin, (2023), Da-Oh et al., (2023), Wulandari et al., (2023), Koç and Sütçü, (2023), Katemba et al., (2022), Malayas et al., (2022), Shyr et al., (2021), Obery et al., (2021), Ghanaat Pisheh et al., (2019), Baguio et al., (2021), Zhu and Urhahne, (2018), Videnovik et al., (2018), Rochimah and Muslim, (2021), Turan and Meral, (2018), Cohn and Fraser, (2016), Barth-Cohen et al., (2016), Xin and Johnson, (2015), Hu and Huang, (2015), Buele-Guerrero and Torres-Carrión, (2022), Siswati et al., (2023), Maming et al., (2023), Cahyani et al., (2022), Suryani and Fauziati, (2022), Can et al., (2021), Anh et al., (2021), Hikmah et al., (2021), Quiroz et al., (2021), Al Munawaroh, (2021), Rahmahani, (2020), Setiyani et al., (2020), Mansur and Fadhilawati, (2019), Barrio et al., (2015), Chien et al., (2015), Agarwal et al., (2014)
Challenges	Binek et al., (2016), Ningsih and Mulyono, (2019), Chien et al., (2015), Naumoska et al., (2022), Suryani and Fauziati, (2022), Janković and Lambić, (2022), Sheard and Chambers, (2014), Lu et al., (2019), Gustafsson and Ryve, (2022), Mahmud and Law, (2022), Sinnivasagam and Hua, (2023), España-Delgado, (2023), Harahap and Kembaren, (2023), Wulandari et al., (2023), Nordin, (2023), Agbatogun, (2014b), Can et al., (2021), Barrio et al., (2015), Mansur and Fadhilawati, (2019), Md Yunus & Azmanuddin bin Azman, (2019), Cahyani et al., (2022), Shyr et al., (2021), Hu and Huang, (2015)

- Kindergarten stage (2–6 years): Children in this phase learn through symbols and images, developing reading comprehension and reasoning with minimal vocabulary. An effective interface includes a blend of pictures, icons, symbols, and simple words for clear communication.
- Concrete operational stage (7–11 years): Children acquire logical thinking skills and comprehend conversations with expanded vocabulary with a mix of pictures and icons in this phase.
- Formal operational stage (12–16 years): Children think hypothetically, demonstrating matured logical thinking and comprehension skills. While pictures, symbols, and icons remain helpful, content should not solely depend on them for communication.

Simplicity, interactivity, durability, and customization should be prioritized in RT's user-friendly interface for school children to create a valuable and constructive tool for involving young learners in the classroom.

- In gamified RTs, students confronted interface issues due to their radiant but overwhelming design. To overcome this issue, vibrant visuals, voice assistance, and text input search may aid children's cognitive recall.
- Researchers pointed out that senior teachers are reluctant to use RT concerning complex functionalities and difficulty in navigation (Sinnivasagam & Hua, 2023). Future RT should be designed to make it easy for teachers with technology and non-technology backgrounds.
- Pilot studies could be conducted with teachers and students of different ages and experiences through individual and focus group interviews for in-depth feedback (Oates et al., 2022). Furthermore, this feedback should also be repetitive after some time for continuous improvement in the design of RT.

Dynamic customization

Students and teachers expressed that future RTs could implement customized features for better learning outcomes, such as breakout rooms for groups (Stanojević & Randelović, 2018).

- Students could personalize their profiles with customizable avatars that create a perception of individuality and engagement (Zainuddin et al., 2020). Moreover, it is also favorable to customize theme options and color schemes that make the RTs more engaging and age-appropriate for school students.
- Introducing a range of questions, including open text, etc., could also be more robust for processing the student responses with large text (Gustafsson & Ryve, 2022). Teachers can adapt the instructional method based on the feedback received from the students through open-text questions.
- It would also be beneficial to implement adaptive questioning based on student responses, and these features may tailor subsequent questions based on student responses. It may also be worth to enable teachers to customize scoring mechanisms for different question types (España-Delgado, 2023).
- The introduction of post-activity rewards, such as virtual prizes or recognition, may serve as incentives for continued participation (Da-Oh et al., 2023).
- RTs could implement the breakout groups or rooms feature to facilitate collaborative learning (Stanojević & Randelović, 2018), allowing students to engage in smaller, focused discussions and activities.

These technological advancements should be associated with educational objectives that support learning rather than a distraction for school children at an early age.

Advanced data protection protocols

Students expressed that RTs often supported anonymous participation, which increased the participation rate, but they were afraid that their recognition might be compromised (Stanojević & Randelović, 2018). The following steps could improve privacy and security.

- All data collection and processing features follow legal requirements and comply with rigorous privacy laws, such as the children's online privacy protection rule.
- Security audits must be conducted recurrently to find and fix vulnerabilities swiftly with robust encryption methods because they reduce the risk of unwanted access by including an extra layer of security for student data (during transmission and storage).
- Open privacy guidelines that support how student data can be processed must be established, and parental consent should be mandatory.

Cross-platform adaptability

RTs confronted compatibility and adaptability challenges that substantially eased their efficiency across various academic levels. Other educational platforms (such as Learning Management System, Moodle, and others) lack support for the RTs (Sun & Hsieh, 2018; Zainuddin et al., 2020). It could be beneficial to implement application programming interfaces to allow the RTs to integrate with existing educational platforms, which can also be valuable to reduce the cost.

Recommendations for teachers

Teachers could focus on the following recommendations that can promote a dynamic, student-centered, and engaging learning environment within the school.

Digital competence

A teacher's technological integration into the classroom depends on the contribution of various factors, such as their knowledge, skills, and attitudes toward technology (Agbatogun, 2014b). There is a positive correlation between digital literacy and performance (Mekota & Marada, 2020; Md Yunus & Azmanuddin bin Azman, 2019). Teachers could participate in workshops focusing on effectively using RTs to enhance classroom engagement and learning. Many of these tools are easy for teachers to use, do not take long to learn, and can have great effects on student learning.

Lecture planning

Teachers should plan the number of questions, question formats, high-quality questions, and pedagogical methods for interactive lectures (Agbatogun, 2014b; Cakiroglu et al., 2018). Preplanning these elements can contribute to a fast pace of lectures and enable teachers to evaluate students' grasp of important concepts (S. Wulandari et al., 2023). Before the actual implementation in the classroom, teachers could conduct a pilot study with some students to overcome the limitations before actual implementation.

Peer collaboration

Teachers might collaborate with peers, sharing effective pedagogical methods for RTs and gaining new insights that advance opportunities for self-reflection. The results of the practices can be used as data to advocate for the purchase of RTs by institutions.

Adaptive lectures

Based on the student's responses, the teacher can either proceed with the lecture, revisit or reteach certain concepts, or plan for additional review in upcoming lectures (Gustafsson & Ryve, 2022). It could be important to instruct the students to use their devices during lessons when questions are being asked through RT. Students could place their devices on their tables and only use them when the teacher asks.

Integration with lesson plans

Teachers expressed that RTs should include in lesson planning through technological, pedagogical, and content knowledge expertise (Sukowati & Sartono, 2020; Nordin, 2023). It could also be possible for teachers to adopt the best pedagogical practices after students' personalized feedback based on students' understanding (Pérez-Segura et al., 2022). Teachers can use RTs to highlight key areas in terms of content through interactive questioning that could help reinforce key concepts by reviewing the student's prior or current knowledge.

Recommendations for Institutions

The following recommendations could be valuable for school leaders to support teachers and students in integrating RTs into teaching and learning practices.

Allocate resources

Teachers get limited support from their institutions to integrate RTs into their pedagogical approaches due to the lack of resources (Agbatogun, 2014b). The lack of resources includes limited staff, lack of proper infrastructure, and the cost of devices or subscription of RTs (Cohn & Fraser, 2016; Xin & Johnson, 2015). Schools should allocate financial resources to support such practices to enhance the classroom environment.

Support interactive practices

Teachers pointed out that without full support from school leaders, teachers achieved limited success in incorporating RTs into their teaching practices (Ningsih & Mulyono, 2019). Enhancing student learning using RTs could not be sufficient to equip the teachers with the tools, but it could also encourage them to experiment with new pedagogical practices.

Impact monitoring

It would be better to monitor and evaluate the use of RTs, especially given the high costs involved, which could lead to necessary adjustments. A well-structured monitoring and evaluation framework could serve as a valuable tool for advocacy of resource allocation to RTs.

Incentives for teachers

Incentives encourage teachers to adopt technology, while support helps them overcome barriers, and perceived benefits motivate continued use. Since the implantation of interactive lectures requires significant time and effort from teachers, institutions often offer

incentives and recognition to acknowledge and appreciate their dedication. Furthermore, specific percentages should also be included in annual reports and promotions to boost the integration of RTs for teaching.

Recommendations for policymakers

Policymakers are also one of the important stakeholders that could implement the following recommendations that can improve the educational landscape countrywide.

Training programs

Various studies reported that teachers' and students' lack of digital literacy is a main barrier to the integration of RTs in classrooms (Janković & Lambić, 2022; Wong & Yunus, 2020; Md Yunus & Azmanuddin bin Azman, 2019) and teachers also showed a willingness to learn new teaching methods (Mahmud & Law, 2022; Nordin, 2023). Policymakers could introduce basic courses in school curricula, training programs, or workshops to ensure students and teachers develop essential skills to navigate digital tools such as RTs.

Equitable access

Teachers and students pointed out that the digital divide is another barrier that hindered the RT integration due to a lack of proper infrastructure (such as low internet connection; Kartini & Sriyanto, 2023; Shyr et al., 2021) and the high cost of devices, especially in developing countries (Setiawan & Soeharto, 2020; Md Yunus & Azmanuddin bin Azman, 2019). It is a global challenge that requires more attention from governments and non-profit organizations. Policymakers could plan policies to ensure the widespread availability of digital devices or mobile phones with subsidized internet services for more equitable access.

Funding and resources

Some schools hesitate to implement RTs in classrooms due to a lack of financial incentives, grants, or tax benefits to schools. It could be vital to allocate resources for infrastructural development, teacher training programs, and technology accessibility, ensuring equitable access nationwide (Agbatogun, 2014b; Cohn & Fraser, 2016). Policies could be designed to encourage school leaders to motivate teachers to integrate RTs with active learning practices.

Clear guidelines for implementation

Policymakers could design clear policies and guidelines on effectively using RTs in various educational settings.

- Ensure that RT integration supports curriculum standards to enhance student engagement, student–teacher interaction, and classroom dynamics that could lead to improved learning outcomes (Katemba et al., 2022).

- Develop guidelines for the design and development of RTs that support diverse learning needs, including students with disabilities, cognitive, vision, or hearing-related issues (Watson & Berry, 2022; Zhu & Urhahne, 2018).
- Encourage collaboration between school leaders, teachers, parents, students, and technologists to foster best practices for RT implementation.

Conclusion

This article addresses the gap in the literature by casting novel insights into the utilization of RTs across several grades in school classrooms. RTs have been used extensively in school classrooms during the last decade. The literature specifies their applicability across diverse courses from small to large classrooms, showcasing an increasing interest in and utilization of these technologies for educational purposes. Furthermore, the findings emphasize the significant contributions of RTs in the school classroom environment, particularly in pedagogical and learning performance, interaction, and feedback. However, conflicting results exist in the literature, with some articles showing a significant difference between RTs-supported and traditional groups, while others indicate no significant difference. While most articles in this SLR supported that RTs are well-acknowledged by teachers and students, offering benefits like enhanced attention and quick feedback, the impact on academic performance varies. In contrast, others emphasized that the overall advantages of RTs outweigh the associated costs and time, underlining the diverse perspectives within educational technology research on the effectiveness of RTs. Moreover, varying opinions on the impact of RTs on the classroom atmosphere emphasize the need for further research to ascertain their effects, as reflected in the increasing number of articles in recent years. While RTs offer significant advantages, their effectiveness is closely tied to the pedagogical methods. Integrating RTs with appropriate pedagogical practices into classrooms is crucial for achieving desired outcomes. This study recommended that teachers could develop digital competence, plan lectures effectively, foster peer collaboration, and implement adaptive teaching methods. Institutions could focus on resource allocation, support interactive teaching practices, monitor impact, and provide incentives for teachers. These recommendations can lead to higher student engagement, increased learning motivation, better learning outcomes, and reduced barriers.

More research is needed to explore the integration of RTs into teaching, addressing the optimal number of questions, deployment timing within the courses, and the most effective teaching methods. Integrating RTs with learning/teaching methods requires further exploration, urging researchers to analyze this integration in-depth. Additionally, the optimal stage of a course for RT implementation needs more investigation, which encourages experimental research in this area. This study contributes a comprehensive overview of RT literature, identifies gaps, and guides future research by consolidating the benefits and challenges of RTs. The study identifies significant gaps in RT literature, offering valuable insights for future research. Future research addressing these gaps will contribute significantly to advancing RT literature. This study focused on articles from a

limited database; potential limitations include excluding gray literature, subjective synthesis, and publication bias.

Author contribution

TMA is the main author and drafted the initial version of the manuscript. TMA and GAS designed the review and formulated the search strategy. TMA and ÖÖ conducted study selection, data extraction, and synthesis. GAS and ÖÖ assisted in editing the manuscript. All authors reviewed, revised, and approved the final manuscript.

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